Daily Briefing - October 25, 2026

Your Daily Tech & Programming Digest

Sunday, October 25, 2026

1000	117883	1166	36
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Peripheral pathway rerouting shapes whole-brain functional network

📖 1 min 📝 25 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Jin-Jing Hu, Yun-Ting Xiang, Bei-Bei Huo, Yi-Fei Zuo, Chen-Yi Peng, He-Wang, Jia-Jia Wu, Mou-Xiong Zheng, Jian-Guang Xu

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https://www.sciencedirect.com/science/article/pii/S0306452226006950?dgcid=rss_sd_all

Corrigendum to “Acupuncture combined with repetitive transcranial magnetic stimulation vs repetitive transcranial magnetic stimulation alone for recovery of hemiparesis after ischemic stroke: A systematic review and meta-analysis” [Neuroscience 616 (2026) 73–91]

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Geng Li, Mengqi Yue, Shuochen Fan, Qichen Bai, Shuai Li, Xiaolin Yang, Xu Chen, Yihao Zhou, Jing Zhong, Jing Shi

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https://www.sciencedirect.com/science/article/pii/S030645222600731?dgcid=rss_sd_all

Polysaccharide-based drug conjugates and related delivery systems for Alzheimer’s disease: from molecular design to translational challenges

📖 1 min 📝 24 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Ngoc Hong Nguyen, Huong Thuy Le, Nhat Thang Thi Nguyen, Thuy Trang Nguyen, Giau Van Vo

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Partial convergence of alternative polyadenylation changes across ALS-Associated RNA-Binding proteins

📖 1 min 📝 10 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Changhe Ji

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https://www.sciencedirect.com/science/article/pii/S0306452226006743?dgcid=rss_sd_all

Genetic variants in NEF2L2 rs35652124 and GABRB3 rs2081648 genes: association with autism spectrum disorder in Iraqi children

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zainab Mohammed Burhan, Wisam Hindawi Hoidy

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https://www.sciencedirect.com/science/article/pii/S0306452226006895?dgcid=rss_sd_all

Propofol suppresses intrinsic excitability and alters excitatory synaptic plasticity in ventral tegmental area dopamine neurons

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Min Hee Lee, Dong Kwan Kim

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https://www.sciencedirect.com/science/article/pii/S0306452226006986?dgcid=rss_sd_all

Astrocytic ablation of truncated TrkB receptors reduces amyloid plaques and prevents synaptic loss in 5xFAD mice

👤 Bass, R., Masterson, H., Chen, Y., Xu, B. 📅 2026-10-24 📖 1 min 📝 234 words

BIORXIV NEUROSCIENCE

Summary: Brain-derived neurotrophic factor (BDNF) is essential for nervous system development and function and exerts its effects primarily through the tropomyosin receptor kinase B (TrkB). The Ntrk2 gene encodes a full-length receptor (TrkB.FL) and two truncated isoforms (TrkB.T1 and TrkB.T2, collectively T...

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https://www.biorxiv.org/content/10.64898/2026.10.20.760443v1?rss=1

HINT1 in ventral hippocampal GABAergic neurons mediates depressive-like behaviors

📅 2026-10-25 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41380-026-03941-5

Integrated microbiome and metabolomic analysis reveals associations between vaginal microbiota composition and psychological profiles

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Disrupted glial-mediated synaptic refinement in Fragile X syndrome

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Emergence of diffusion models from associative memory through memorization and generalization

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Effects of vertebral shortening during corrective osteotomy for spinal deformity on spinal cord biomechanics and cerebrospinal fluid-anterior spinal artery fluid-structure interaction

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Single-cell transcriptomic insights into conserved and divergent oligodendrocyte lineage programs across species

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https://www.nature.com/articles/s42003-026-11256-y

SIK3 controls synaptic localization of NMDA receptors by phosphorylating GluN1

Thomas C. Südhof 2026-10-21 1 min 41 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 21 October 2026; doi:10.1038/s41593-026-02469-9 SIK3 kinase controls synaptic NMDA receptor recruitment by phosphorylating GluN1 at a single site. Loss or inhibition of SIK3 markedly reduces...

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https://www.nature.com/articles/s41593-026-02469-9

A corticohypothalamic circuit that mediates maternal awakening to pup cries

Feixue Liang 2026-10-21 1 min 45 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 21 October 2026; doi:10.1038/s41593-026-02458-y Tang, Wu et al. identify a brain circuit that enables mother mice to wake selectively to pup cries. Motherhood strengthens this pathway throug...

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https://www.nature.com/articles/s41593-026-02458-y

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob 2026-10-13 1 min 229 words FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1778980

Wave-function formalism for evolving neural dynamics: a mathematical framework informed by EEG

Madelyn Klebba 2026-10-12 1 min 268 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: The organization of distributed neural activity remains one of the central challenges in computational neuroscience. Contemporary frameworks, including Integrated Information Theory, Global Workspace Theory, entropy-based approaches, and electromagnetic field theories, have advanced our understandin...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1812909

Implementing the ILAE 2025 seizure taxonomy in medical software

Jukka Peltola 2026-10-13 1 min 241 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: The International League Against Epilepsy (ILAE), working via a multidisciplinary task force, has issued a position paper with updated seizure classification guidelines (2025) replacing its previous guidance (2017), establishing a revised de-facto standard for clinical assessment and classification ...

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Correction: Arts engagement, active inference, and allostatic regulation: a neurobiological framework for understanding the health effects of aesthetic experience

Pier Luigi Sacco 2026-10-13 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2004139

Generative-augmented multimodal neuroaffective decoding for product evaluation: toward neurofeedback-driven human-in-the-loop co-design

👤 Ming Deng 📅 2026-10-21 📖 1 min 📝 227 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundKansei engineering traditionally relies on self-reported evaluations of product impressions, which are susceptible to contextual influences, recall bias, and assessment timing. Multimodal psychophysiological measurements offer complementary insights into affective responses, but integratin...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1997654

From a surgeon's stroke to a third hemisphere—The auxilium architecture for brain restoration and extension

👤 Mijail Demian Serruya 📅 2026-10-21 📖 1 min 📝 344 words FRONTIERS HUMAN NEUROSCIENCE

Summary: A surgeon was exercising on a stationary bicycle when he had a large ischemic left middle cerebral artery territory stroke, complicated by hemorrhagic transformation requiring hemicraniectomy and cranioplasty. He developed right-sided spastic hemiplegia and a mixed aphasia. This paper takes that cas...

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Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI

👤 Christoffer Olsson 📅 2026-10-21 📖 1 min 📝 245 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionPredictions of the electric field (E-field) induced by transcranial direct current stimulation (tDCS) depend on how brain conductivity is modeled. White matter anisotropy is conventionally estimated from single-shell diffusion tensor imaging (DTI). Multidimensional diffusion MRI (MD-dMRI...

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Effectiveness of bimanual kinesthetic teaching through different controller settings

👤 Andrej Gams 📅 2026-10-21 📖 1 min 📝 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

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Role of astrocytes in obesity-induced hypertension

Madhan Subramanian 2026-10-21 1 min 223 words FRONTIERS NEUROSCIENCE

Summary: Although obesity is a significant contributor to hypertension and cardiovascular disease, the mechanisms that connect excess adiposity to persistently elevated blood pressure extend far beyond peripheral metabolic and vascular changes. An increasing body of evidence suggests that a substantial compo...

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Gene expression analysis of Fmr1-KO mouse caudate putamen reveals new preclinical biomarkers for fragile X syndrome

Manuela Prokesch 2026-10-23 1 min 248 words FRONTIERS NEUROSCIENCE

Summary: BackgroundFragile X syndrome is a monogenic cause of autism spectrum disorder. Fmr1-KO mice are a well-known model for the disease, and the model's behavioral phenotype has been well-characterized. Unfortunately, translational molecular tissue biomarkers for this model remain sparse.MethodsTo identi...

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Ultra high-density, 4096-channel intraoperative neurophysiological brain mapping for functional localization of the human central sulcus

👤 Alexandra Fink Skular, Gelana Tostaeva, Elton Ho, Jacqueline Dister, Daniel D Cummins, Katrina Barth, Lizbeth Nunez, Adam Poole, Craig H Mermel, Benjamin Rapoport, Joshua B Bederson and Ignacio Saez

📅 2026-10-20 📖 1 min 📝 378 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Intraoperative localization of the central sulcus (CS) using somatosensory evoked-potential (SSEP) phase reversal (PR) is routinely performed with sparse subdural strip electrodes. We evaluated whether a multi-array, 4096-channel surface micro-electrocorticography (μECoG) configuration co...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Screening-based prevalence and axis distribution of astigmatism among children and adolescents in Lintong District, Xi'an: a large-scale school-based cross-sectional study

👤 Zhaojiang Du
📅 2026-10-23 📖 1 min 📝 51 words

LOW VISION

Summary: CONCLUSIONS: Astigmatism was common in this large district school screening dataset and differed by school stage, axis type, and refractive-status category. The findings may help local screening programs plan re-examination and referral pathways, but they should not be interpreted as cycloplegic cli...

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Genome mining and comparative genome analysis of Pseudomonas fluorescens revealed diverse biosynthetic and metabolic potential

👤 Farida Begum 📅 2026-10-24 📖 1 min 📝 70 words LOW VISION

Summary: Bacterial genome sequencing has been used to identify a myriad of bioactive compounds that have yet to be characterised. Pseudomonas fluorescens is a widely distributed Gram-negative bacterium known for its plant growth-promoting traits and production of diverse secondary metabolites. P. fluorescens...

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Clinical profiles, referral patterns and low vision aid prescribing across four rehabilitation centres in Ghana: protocol for a multicentre retrospective record review

Werner Eisenbarth 2026-10-24 1 min 79 words LOW VISION

Summary: BACKGROUND: Low vision, defined as best-corrected visual acuity of less than 6/18 in the better eye or a visual field constriction of 20° or less, is a condition in which residual vision may be optimized through optical and non-optical aids. Despite the importance of low vision services in maximizin...

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Retinal Artificial Intelligence Reclassifies SCORE2-Based Cardiovascular Risk Across the ESC/EAS Statin-Indication Boundary: A UK Biobank Prospective Cohort Study

Sunjin Hwang 2026-10-24 1 min 28 words LOW VISION

Summary: CONCLUSION: Adding the retinal AI risk score to SCORE2 produced modest bidirectional risk reclassification across the ESC/EAS statin-indication boundary; these findings warrant prospective validation.Abstract word count: 250 words.

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Long-Term Visual, Refractive, and Patient-Reported Satisfaction of the Tel-Aviv Protocol (ePRK-CXL) for Progressive Keratoconus

👤 Igor Kaiserman 📅 2026-10-24 📖 1 min 📝 41 words LOW VISION

Summary: CONCLUSIONS: At a mean follow-up of ~8 years, the Tel-Aviv Protocol provided durable visual and refractive improvement, stabilized keratometry without progression, and yielded high patient satisfaction with low symptom burden. These results support ePRK-CXL as a tissue-sparing option for progressive...

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Clinical significance of "complex" multimorbidity in hospitalised adult patients

👤 Gino Roberto Corazza 📅 2026-10-24 📖 1 min 📝 65 words LOW VISION

Summary: The term "complex multimorbidity" lacks universal consensus and its relevance remains unclear. We aimed to assess the prevalence of complex multimorbidity and its association with major healthcare outcomes. This was a prospective, single-centre study (internal medicine ward). We collected sociodemog...

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Tele-optometry for eye disease screening: a scoping review of context-dependent diagnostic performance and limitations for large-scale implementation

Chigozie Ekenze 2026-10-24 1 min 50 words LOW VISION

Summary: CONCLUSION: Tele-diagnostic screening shows promise for high-prevalence conditions with straightforward criteria, such as DR and refractive error, but is less reliable for complex diseases like glaucoma. These conclusions are constrained by limited downstream-outcome reporting and geographic concent...

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Early changes in the central vision and the macula in diabetic retinal disease: associations with serum biomarkers

Yoko Ozawa 2026-10-24 1 min 62 words LOW VISION

Summary: CONCLUSION: This study links subtle macular neural and vascular changes with systemic biomarkers in diabetic patients without visible macular lesions. Impaired central vision was associated with shorter PROS, lower RPE volume, and reduced vascular parameters. These findings improve understanding of ...

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From task-specific AI to multimodal foundation models in PET/CT: a readiness framework for clinical translation

👤 Jia Wu 📅 2026-10-24 📖 1 min 📝 59 words LOW VISION

Summary: Artificial intelligence in PET/CT is evolving from narrowly optimized systems toward reusable pretrained representations and multimodal models. PET/CT is a demanding test case because clinically meaningful interpretation must preserve quantitative molecular information while integrating anatomic CT,...

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Diagnosis-Related Group Payment Systems and Hospital Efficiency, Quality, and Coding Behavior: A Systematic Review and Narrative Synthesis

👤 Wael Alsebaei 📅 2026-10-24 📖 1 min 📝 66 words LOW VISION

Summary: Diagnosis-related group (DRG)-based payment systems are used globally to incentivize hospital efficiency, yet their impact on clinical quality and documentation behavior remains variable. As Saudi Arabia implements the Australian Refined DRGs (AR-DRG) framework under Vision 2030, a synthesis of inte...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

Peng Chen 2026-10-23 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

Xiaojun Zhou 2026-10-23 1 min 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

Hussein Youssef 2026-10-23 1 min 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

Leticia González 2026-10-23 1 min 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Nonpharmacological Interventions in Degenerative Cerebellar Ataxia: A Systematic Review

Pramod Kumar Pal 2026-10-23 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: NPI may confer short-term symptomatic and functional benefits in degenerative cerebellar ataxia, particularly in axial motor domains, supported by the largest evidence for structured rehabilitation followed by neuromodulation, although heterogeneous. Future multicenter, biomarker-inform...

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Effects of Orbitofrontal Cortex tDCS on Cognitive Control in Individuals With Obesity: A Randomized Sham-Controlled Trial

👤 Stephanie L Borgland
📅 2026-10-23 📖 1 min 📝 43 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Unlike what has been reported in lean individuals, a single session of OFC-targeted tDCS is insufficient to modify cognitive control in individuals with obesity. However, the cognitive effects of a single session of OFC stimulation depend on individual neurocognitive characteristics in ...

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Disrupting objectification: transcranial Direct Current Stimulation (tDCS) enhances the inversion effect toward sexualized women

Arwen Stephens
2026-10-24 1 min 68 words TDCS TACS TRNS

Summary: Sexual objectification is associated with viewing women in an object-like manner, often indexed perceptually by a reduced inversion effect for sexualized women relative to men. Prior research has shown that this index can be attenuated through social manipulations; here we investigated whether it ca...

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Efficacy and Safety of Perioperative Transcranial Electrical Stimulation for Postoperative Pain and Related Outcomes: A Systematic Review and Meta-Analysis With Trial Sequential and Subgroup Analyses

👤 Qing Zhong
📅 2026-10-24 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Current evidence suggests that tES, as an adjuvant perioperative intervention, may alleviate postoperative pain, reduce opioid consumption, and decrease the incidence of postoperative delirium. However, the effect of tES on postoperative quality of recovery remains unclear. Considering ...

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Sensory cortical disinhibition is required for long-term fear memory in humans

👤 Wen Li 📅 2026-10-24 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: The transformation of transient threats into enduring memories is fundamental to survival, yet how the human brain constructs long-term fear memory remains elusive. Early sensory cortex, traditionally viewed as a passive feature analyzer, is increasingly recognized as an active participant in fear p...

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Neuromodulatory interventions for Alzheimer's disease

👤 Wandong Zhang 📅 2026-10-24 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: Alzheimer's disease (AD) and mild cognitive impairment (MCI) represent an escalating global public health challenge with major socioeconomic consequences. Although anti-amyloid monoclonal antibodies have introduced disease-modifying treatment options for early symptomatic AD, their benefits remain m...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

👤 Balaji Jayaprakash 📅 2026-10-22 📖 1 min 📝 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

👤 Jing Zhang 📅 2026-10-23 📖 1 min 📝 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

👤 Rihui Li 📅 2026-10-23 📖 1 min 📝 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

Lulu Song 2026-10-24 1 min 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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How School-Based Interventions Shape the Developing Brain during 3-12 Years: A Systematic Review and Exploratory Meta-analysis

👤 Hui Li 📅 2026-10-24 📖 1 min 📝 56 words FNIRS

Summary: The effects of school-based interventions (SBIs) on children's neurodevelopment are poorly synthesized. This systematic review integrates neuroscientific evidence on how structured educational activities influence brain development in children aged 3-12 years. Following PRISMA guidelines, we identif...

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Transformer networks are robust learners of source localization for EEG measurements

👤 Moritz Dannhauer 📅 2026-10-23 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

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Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

Chen Wang 2026-10-23 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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Generalizable neural decoding for clinically scalable brain-computer interfaces

👤 Jie Chen 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces promise restored communication and movement, yet accuracy achieved under development rarely survives new sessions, users, tasks, or hardware. Generalization, not peak performance, determines clinical scalability. Here we examine generalizable neural decoding from developmen...

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Polygenicity shapes distinct brain patterns and clinical profiles of bipolar disorder

👤 Shaohua Hu 📅 2026-10-23 📖 1 min 📝 41 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Our findings reveal associations between polygenicity and distinct BD brain patterns and clinical profiles. We identified ancestry-specific causal variants and bidirectional BD-IDP relationships. PRS associations with depressive symptoms, anxiety, suicide risk, and metabolic phenotypes ...

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Combined neuromodulation techniques via a central-peripheral-central strategy facilitate functional recovery from AIS A to AIS C in a patient with complete spinal cord injury: A case report

Liansheng Zhong
2026-10-24 1 min 48 words BRAIN COMPUTER INTERFACE

Summary: CONTEXT: Complete spinal cord injury (SCI) has traditionally been considered irreversible. However, accumulating evidence suggests that subclinical neural pathways may persist. Multimodal combined neuromodulation targeting brain-spinal-peripheral circuits holds promise to unlock latent neural plasti...

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Case Report: An MDT-guided multimodal protocol for large-area stroke plateau - integrating neuromodulation with multimodal assessment

Xiao-Xia Du 2026-10-24 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this patient with large-area stroke who showed limited response to conventional rehabilitation, an MDT-guided, dynamically adjusted multimodal intervention was temporally associated with concurrent recovery across cognitive, language, motor, and daily living domains. The concomitant n...

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Correction: Perioperative management for invasive brain-computer interface implantation under surgical navigation in a patient with traumatic spinal cord injury: a case report

Wenyao Cui 2026-10-24 1 min 6 words BRAIN COMPUTER INTERFACE

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1892331.].

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Limits of Trial-Adaptive Neural-Language Fusion Across Large Language Models in P300 Brain-Computer Interfaces

Yiftach Barash 2026-10-24 1 min 21 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: A tuned fixed weight is a difficult-to-beat default across the tested scale range; reliability estimation gave no deployable adaptive advantage.

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Creating online cognitive assessment platform for children's literacy and math skills: a case study illustrated with the ACE kids platform

U Maurer 2026-10-24 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: Computerized versions of paper-and-pencil tests made behavioral assessments of children more automatic and precise. The COVID-19 pandemic further demonstrated the need for assessments to be offered online to allow for remote data collection. However, the design and implementation of computerized tes...

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Easily download files from the Open Science Framework with Papercheck

noreply@blogger.com (Daniel Lakens)
2025-07-22 3 min 765 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

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http://ieeexplore.ieee.org/document/11585967

Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

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http://ieeexplore.ieee.org/document/11595768

VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

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http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

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http://ieeexplore.ieee.org/document/11493624

A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11489488

Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

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http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

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Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

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Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

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Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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http://ieeexplore.ieee.org/document/11741005

Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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http://ieeexplore.ieee.org/document/11740942

Therapeutic potential of proline-rich peptide-1 (PRP-1) in counteracting amyloid-β-mediated cortical and hippocampal neurodegeneration in a rat model

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Harutyun Stepanyan, Hamzeh Ghorbani, Mikhail Poghosyan, Margarita Danielyan, Arsen Minasyan, Arsen Hunanyan, Naira Hunanyan

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Decoding spatial attention in the cocktail party problem using whole-head high-density fNIRS

📖 1 min 📝 47 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Sudan Duwadi, De'Ja Rogers, Alex D. Boyd, Laura B. Carlton, Yiwen Zhang, Anna Kawai Gaona, Aneesa Diya Pathiyaparambath, Ravin Chaudhury, Grace Magee, Bernhard B Zimmermann, Walker J O'Brien, Alexander...

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Subjective emotional ratings dissociate brain activation and network patterns during counterfactual outcome processing

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Tingting Zhang, Denghui Bai, Xiaotong Wei, Lei Zhuo, Qiuzhu Zhang, Junjun Zhang, Zhenlan Jin, Ling Li

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Active locomotion predictively rescues head direction attractor dynamics in head-fixed mice

📅 2026-10-25 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41467-026-79031-5

Early lineage divergence segregates sensory and motor-associative identities in the thalamus

👤 Guillermina López-Bendito 📅 2026-10-20 📖 1 min 📝 35 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 20 October 2026; doi:10.1038/s41593-026-02485-9 Sensory and motor-associative identities in the thalamus arise from two early-diverging lineages, with sensory neurons progressing through sha...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02485-9

Dysregulated cholesterol trafficking through ABCA1 drives neurodegeneration in Parkinson’s disease

Lei Cao 2026-10-20 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 20 October 2026; doi:10.1038/s41593-026-02487-7 This study identifies ABCA1 as a driver of dysregulated cholesterol trafficking in Parkinson’s disease, showing that reducing ABCA1 in astrocy...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02487-7

Sustained antidepressant-like effects of psychedelics through 5-HT_{2A}R–β-arrestin signaling

Sheng Wang 2026-10-23 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02497-5 This study shows that the sustained antidepressant-like effect of psychedelics is driven by 5-HT2A receptor–β-arrestin signaling. Repeated low...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02497-5

Identification of candidate markers for the isolation of neuron-specific extracellular vesicles

David R. Walt
2026-10-23 1 min 41 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 23 October 2026; doi:10.1038/s41593-026-02466-y The authors developed methods to identify extracellular vesicles (EVs) in the blood that come from the brain by capturing EV surface proteins....

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02466-y

Neuropharmacology of language: ERP evidence for catecholaminergic modulation of syntactic processing

Yingying TanAshley G. LewisIris C. SchmitsRobert-Jan VerkesRoshan CoolsPeter Hagoortahttps://ror.org/01bn89z48Shanghai International Studies University, Institute of Language Sciences, Shanghai 201620, Chinabhttps://ror.org/00671me87Max Planck Institute for Psycholinguistics, Neurobiology of Language Department, Nijmegen 6525 XD, the Netherlandschttps://ror.org/03q8dnn23City University of Hong Kong, Department of Linguistics and Translation, Kowloon, Hong Kong Special Administrative Region, Chinadhttps://ror.org/016xsfp80Radboud University, Donders Institute for Brain, Cognition and Behaviour, Center for Cognitive Neuroimaging, Nijmegen 6525 EN, the Netherlandsehttps://ror.org/05wg1m734Radboud University Medical Center, Department of Psychiatry, Nijmegen 6525 GA, the Netherlands

📅 2026-10-22 📖 1 min 📝 42 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceHow the brain’s neuromodulatory systems influence language, a defining human capacity, remains poorly understood. Here we show that catecholaminergic neurotransmission—primarily involving dopamine ...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2606806123?af=R

Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao 2026-10-02 1 min 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1971441

A systematic review of machine learning approaches for MRI-based brain tumor analysis

Almira Melsovna Kustubayeva
2026-10-12 1 min 213 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Machine learning is now widely applied to magnetic resonance imaging-based brain tumor analysis, with task-specific performance metrics: the Dice similarity coefficient (DSC) quantifies segmentation overlap, while the area under the curve (AUC) and accuracy measure classification performance. A PRIS...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1937002

Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

Patrick Krauss
2026-10-20 1 min 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1886601

Artificial intelligence-generated and algorithmically personalized music interventions and their effects on brain function, behavior, and health: a critical review

Jingting Jin 2026-10-23 1 min 310 words FRONTIERS HUMAN NEUROSCIENCE

Summary: This critical review synthesizes empirical studies of algorithmically personalized and artificial-intelligence-generated music interventions reporting neurophysiological, neuroimaging, or objective behavioral outcomes. Literature was identified through structured searches of PubMed, Scopus, Google S...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1967909

Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

Sivakumar Balasubramanian 2026-10-20 1 min 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

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Research progress on the mechanism of microglia–astrocyte signal crosstalk in central sensitization of vestibular migraine

Huailiang Zhang
2026-10-23 1 min 127 words

FRONTIERS NEUROSCIENCE

Summary: Vestibular migraine (VM) is a vestibular disorder characterized by recurrent episodes of vertigo, frequently accompanied by headache, nausea, and vomiting, which substantially reduce patients' quality of life. The pathogenesis of VM remains incompletely understood. Current studies mainly focus on io...

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From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan
2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ultrabright Quasi-Compact Quantum Dot Assembly for Sensitive and Color-Fixed Immunoassays

👤 Liang Huang
📅 2026-10-23 📖 1 min 📝 63 words

LOW VISION

Summary: Home-based immunoassay platforms constitute critical tools for the rapid screening of infectious respiratory viruses. Notably, viral loads can remain low at certain stages yet vary over several orders of magnitude during infection, and such variations are closely associated with the disease progress...

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Supercritical Carbon Dioxide-Peracetic Acid Treatment Inactivates Low-Level Bacterial Inocula in Collagen Analogue-Phosphorylcholine Hydrogels while Preserving Functional Properties

👤 May Griffith
📅 2026-10-23 📖 1 min 📝 57 words LOW VISION

Summary: Terminal sterilization of hydrogels is challenging because their high-water-content, chemical structures, and cell-friendly properties are often altered by conventional sterilization methods. Hydrogel implants comprising collagen-like peptide (CLP), polyethylene glycol (PEG), and poly (2-methacryloy...

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A Bidirectional Photosensitive Multimodal PdSe2/MoS2 van der Waals Heterostructure Memristor Array for In-Sensor Neuromorphic Vision

👤 Jun He 📅 2026-10-23 📖 1 min 📝 60 words LOW VISION

Summary: Optoelectronic van der Waals (vdW) memristors integrating photosensing, memory, and computing capabilities are promising for artificial vision but are limited by single photoresponse modes, low yields, and high device variability. Here, we demonstrate a 1-kb PdSe2/MoS2 van der Waals heterostructure ...

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Al-Stoichiometry-Engineered Defect Profiles in InAlZnO Nanofibers for Balanced Charge Transport and Optical Plasticity

👤 Fengyun Wang
📅 2026-10-23 📖 1 min 📝 64 words

LOW VISION

Summary: Metal oxide optoelectronic synapses are promising for neuromorphic vision, but combining a low dark current with sustained photoinduced plasticity remains challenging. Here, Al stoichiometry is engineered in electrospun In1AlxZn0.08O (x = 0.007, 0.011, and 0.015) nanofiber networks to tune sub-bandg...

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Effect of 0.01%, 0.02% and 0.04% Atropine eyedrops vs Placebo on myopia incidence in pre-myopic children: A randomized clinical trial

Xiangui He 2026-10-23 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Low-concentration atropine demonstrated a dose-response relationship for reducing cumulative myopia incidence and slowing refractive progression in pre-myopic children, but no such relationship on axial length. This preliminary evidence suggests potential for early myopia prevention, bu...

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Diagnosis of a chiasmal lesion compressing optic chiasm in patients with severe retinal degeneration: a two-case report

Kyung-Ah Park 2026-10-23 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Optic chiasmal compression should be considered in patients with severe retinal degeneration presenting with progressive visual decline that cannot be fully explained by retinal pathology. Visual field testing and timely neuroimaging, particularly MRI, are essential for accurate diagnos...

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Reading Arabic Under Simulated Low Vision: Orthographic Complexity Shapes the Effect of Contrast Polarity

👤 Mosaad Alhassan
📅 2026-10-23 📖 1 min 📝 45 words

LOW VISION

Summary: CONCLUSION: Arabic relies on small, high-spatial-frequency features such as dots for letter discrimination, which appear to modulate how contrast polarity interacts with optical degradation at threshold. These findings provide preliminary evidence that may guide future studies of reading performance...

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Screening-based prevalence and axis distribution of astigmatism among children and adolescents in Lintong District, Xi'an: a large-scale school-based cross-sectional study

Zhaojiang Du
2026-10-23 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Astigmatism was common in this large district school screening dataset and differed by school stage, axis type, and refractive-status category. The findings may help local screening programs plan re-examination and referral pathways, but they should not be interpreted as cycloplegic cli...

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Genome mining and comparative genome analysis of Pseudomonas fluorescens revealed diverse biosynthetic and metabolic potential

👤 Farida Begum 📅 2026-10-24 📖 1 min 📝 70 words LOW VISION

Summary: Bacterial genome sequencing has been used to identify a myriad of bioactive compounds that have yet to be characterised. Pseudomonas fluorescens is a widely distributed Gram-negative bacterium known for its plant growth-promoting traits and production of diverse secondary metabolites. P. fluorescens...

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Clinical profiles, referral patterns and low vision aid prescribing across four rehabilitation centres in Ghana: protocol for a multicentre retrospective record review

Werner Eisenbarth 2026-10-24 1 min 79 words LOW VISION

Summary: BACKGROUND: Low vision, defined as best-corrected visual acuity of less than 6/18 in the better eye or a visual field constriction of 20° or less, is a condition in which residual vision may be optimized through optical and non-optical aids. Despite the importance of low vision services in maximizin...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

Xin Zhou 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

Peng Chen 2026-10-23 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

Xiaojun Zhou 2026-10-23 1 min 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

👤 Hussein Youssef 📅 2026-10-23 📖 1 min 📝 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

👤 Leticia González 📅 2026-10-23 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Nonpharmacological Interventions in Degenerative Cerebellar Ataxia: A Systematic Review

👤 Pramod Kumar Pal 📅 2026-10-23 📖 1 min 📝 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: NPI may confer short-term symptomatic and functional benefits in degenerative cerebellar ataxia, particularly in axial motor domains, supported by the largest evidence for structured rehabilitation followed by neuromodulation, although heterogeneous. Future multicenter, biomarker-inform...

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Effects of Orbitofrontal Cortex tDCS on Cognitive Control in Individuals With Obesity: A Randomized Sham-Controlled Trial

👤 Stephanie L Borgland 📅 2026-10-23 📖 1 min 📝 43 words TDCS TACS TRNS

Summary: CONCLUSIONS: Unlike what has been reported in lean individuals, a single session of OFC-targeted tDCS is insufficient to modify cognitive control in individuals with obesity. However, the cognitive effects of a single session of OFC stimulation depend on individual neurocognitive characteristics in ...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi 📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

Rihui Li 2026-10-23 1 min 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

Georg S Kranz 2026-10-23 1 min 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

Lulu Song 2026-10-24 1 min 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

Qingguo Wei 2026-10-23 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

👤 Jiayuan He 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Training Free Signal Enhancement by Random Sampling for Steady State Visual Evoked Potential BCI

👤 Yoshikazu Washizawa 📅 2026-10-23 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: The steady-state visual evoked potential (SSVEP) is widely used for
electroencephalography (EEG)
based brain computer interfaces (BCIs).
In order to determine the target commands for SSVEP-BCIs,
deep neural networks have been widely used.
However, learning-based approaches requir...

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Decoding attempted handwritten characters based on temporal endovascular neural features

Peter E Yoo 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: We evaluated temporal neural features for motor decoding using a minimally-invasive endovascular brain-computer interfaces (BCI). We hypothesised that relatively long compound movements generate distinct patterns of neural activity that are encoded by temporal features.

Approach. A single p...

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Transformer networks are robust learners of source localization for EEG measurements

Moritz Dannhauer 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

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Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

Chen Wang 2026-10-23 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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Generalizable neural decoding for clinically scalable brain-computer interfaces

👤 Jie Chen 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces promise restored communication and movement, yet accuracy achieved under development rarely survives new sessions, users, tasks, or hardware. Generalization, not peak performance, determines clinical scalability. Here we examine generalizable neural decoding from developmen...

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Polygenicity shapes distinct brain patterns and clinical profiles of bipolar disorder

👤 Shaohua Hu 📅 2026-10-23 📖 1 min 📝 41 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Our findings reveal associations between polygenicity and distinct BD brain patterns and clinical profiles. We identified ancestry-specific causal variants and bidirectional BD-IDP relationships. PRS associations with depressive symptoms, anxiety, suicide risk, and metabolic phenotypes ...

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The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

📅 2024-01-22 📖 1 min 📝 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition's foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

Sunday Daily Thread: What's everyone working on this week?

👤 /u/ AutoModerator 📅 2026-10-25 📖 1 min 📝 211 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: What's Everyone Working On This Week? 🛠️</h1> <p>Hello r/Python! It's time to share what you've been working on! Whether it's a work-in-progress, a completed masterpiece, or just a rough idea, let us know what you're up to!</p...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xennpj/sunday_daily_thread_whats_everyone_working_on/

America has lost 200k of jobs in media, film and the arts. How much is AI?

👤 gnabgib 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.apricitas.io/p/ai-and-the-fall-of-the-creative-class Comments URL: https://news.ycombinator.com/item?id=50226419 Points...

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https://www.apricitas.io/p/ai-and-the-fall-of-the-creative-class

Why Gopher in 2026?

👤 andsoitis 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://gopher.muffinlabs.com/sdf.org/users/nervtoji/phlog/001.txt Comments URL: https://news.ycombinator.com/item?id=50226577 Po...

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https://gopher.muffinlabs.com/sdf.org/users/nervtoji/phlog/001.txt

FindMyWeb: Apple Find My, reverse enginiered

👤 Velocifyer 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://codeberg.org/Misom/FindMyWeb Comments URL: https://news.ycombinator.com/item?id=50226646 Points: 27 # Comments: 7

🔗 **Read full article:**
https://codeberg.org/Misom/FindMyWeb

Cerebras Deploys Trillion-Parameter AI Models

👤 gmays 📅 2026-10-25 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.cerebras.ai/blog/how-we-deploy-trillion-parameter-frontier-models-on-cerebras Comments URL: <a href="https://news.ycombinator.com/item?id=50227356">https://news....

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https://www.cerebras.ai/blog/how-we-deploy-trillion-parameter-frontier-models-on-cerebras

Meta employees say Lula’s accounts restricted in Brazil election: Report

👤 nidnogg 📅 2026-10-25 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.aljazeera.com/economy/2026/10/16/meta-employees-say-lulas-accounts-restricted-in-brazil-election-report Comments URL: <a href="https://...

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https://www.aljazeera.com/economy/2026/10/16/meta-employees-say-lulas-accounts-restricted-in-brazil-election-report

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

Adriel Carridice 2025-11-04 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Single-cell RNA sequencing reveals vascular heterogeneity and cell–cell communication remodeling in the glioma blood–tumor barrier

1 min 31 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Zhipeng Zhao, Jiachen Lang, Haiyan Gu, Luodan Wang, Jianchang Luo, Jiawang Lang, Qiongying Yao, Xinyu Qiao, Zida Wang, Ling Yin, Caidi He

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004579?dgcid=rss_sd_all

From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

Christine Fernandez-Maloigne
2026-10-01 1 min 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1790241

Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka
2026-10-23 1 min 318 words

FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1875377

Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

Chi Chung Alan Fung 2026-10-23 1 min 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1923602

Orientation detection for self-calibrating neuromorphic systems

Elisabetta Chicca 2026-10-23 1 min 183 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Bioinspired sensory architectures help build adaptive touch systems with efficient hardware implementations. Orientation detection is a fundamental capability for manipulating objects in real-world scenarios, with diverse applications in robotic manipulation and manufacturing assessment processes. I...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1878557

A thalamus-centered network model of restless legs syndrome: an integrative hypothesis illustrated by three clinical phenotypes

Szabolcs Várbíró 2026-10-23 1 min 231 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Restless legs syndrome (RLS) is a clinically heterogeneous sensorimotor disorder whose pathophysiology reflects interacting biological mechanisms rather than a single causal pathway. Established influences—including brain iron biology, dopaminergic regulation, glutamatergic and GABAergic balance, ad...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1977941

A systems-level energetic framework for conscious processing: energetic organization as a constraint on large-scale neural integration

Holly Pollard-Wright 2026-10-23 1 min 259 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conscious processing is associated with coordinated interactions among distributed neural systems, yet the biological constraints that enable and sustain large-scale neural integration remain incompletely understood. Contemporary theories emphasize computational, informational, and network mechanism...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966072

Cortical responses to lipread and text-based audiovisual stimuli in adults with and without dyslexia

Milene Bonte 📅 2026-10-23 📖 1 min 📝 235 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Phonetic recalibration enables listeners to adjust their perception of ambiguous speech sounds based on visual contextual information. In line with a specific difficulty in forming letter-speech sound associations, adults with dyslexia have been found to show no text-based, but intact lipread-based,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1806964

Editorial: New horizons in stroke management, volume II

Nasser Kashou 📅 2026-10-23 📖 1 min 📝 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2023091

Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1985500

The perioperative sensory–cognitive interface: multimodal neuroimaging of hierarchical network reconfiguration in perioperative neurocognitive disorders

Zhiheng Liu 2026-10-21 1 min 257 words FRONTIERS NEUROSCIENCE

Summary: Perioperative neurocognitive disorders (PNDs) are commonly attributed to neuroinflammation, cerebrovascular stress, anesthetic neurophysiology, and pre-existing brain vulnerability. These mechanisms identify important insults but do not fully explain how they become disturbances of attention, percep...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1956385

Comparison of short-term peripheral nerve stimulation, single-target supraorbital nerve pulsed radiofrequency, and dual-target supraorbital nerve–stellate ganglion pulsed radiofrequency for herpes zoster ophthalmicus–associated pain: a retrospective cohort study

Wei Wang 2026-10-23 1 min 326 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveTo compare the clinical outcomes of short-term peripheral nerve stimulation (st-PNS), supraorbital nerve pulsed radiofrequency (SON-PRF), and combined supraorbital nerve and stellate ganglion pulsed radiofrequency (SON + SG-PRF) for herpes zoster ophthalmicus (HZO)-associated pain.MethodsTh...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1976324

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261024193344&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ultrabright Quasi-Compact Quantum Dot Assembly for Sensitive and Color-Fixed Immunoassays

👤 Liang Huang 📅 2026-10-23 📖 1 min 📝 63 words LOW VISION

Summary: Home-based immunoassay platforms constitute critical tools for the rapid screening of infectious respiratory viruses. Notably, viral loads can remain low at certain stages yet vary over several orders of magnitude during infection, and such variations are closely associated with the disease progress...

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Supercritical Carbon Dioxide-Peracetic Acid Treatment Inactivates Low-Level Bacterial Inocula in Collagen Analogue-Phosphorylcholine Hydrogels while Preserving Functional Properties

👤 May Griffith
📅 2026-10-23 📖 1 min 📝 57 words

LOW VISION

Summary: Terminal sterilization of hydrogels is challenging because their high-water-content, chemical structures, and cell-friendly properties are often altered by conventional sterilization methods. Hydrogel implants comprising collagen-like peptide (CLP), polyethylene glycol (PEG), and poly (2-methacryloy...

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A Bidirectional Photosensitive Multimodal PdSe2/MoS2 van der Waals Heterostructure Memristor Array for In-Sensor Neuromorphic Vision

👤 Jun He 📅 2026-10-23 📖 1 min 📝 60 words LOW VISION

Summary: Optoelectronic van der Waals (vdW) memristors integrating photosensing, memory, and computing capabilities are promising for artificial vision but are limited by single photoresponse modes, low yields, and high device variability. Here, we demonstrate a 1-kb PdSe2/MoS2 van der Waals heterostructure ...

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Al-Stoichiometry-Engineered Defect Profiles in InAlZnO Nanofibers for Balanced Charge Transport and Optical Plasticity

👤 Fengyun Wang
📅 2026-10-23 📖 1 min 📝 64 words

LOW VISION

Summary: Metal oxide optoelectronic synapses are promising for neuromorphic vision, but combining a low dark current with sustained photoinduced plasticity remains challenging. Here, Al stoichiometry is engineered in electrospun In1AlxZn0.08O (x = 0.007, 0.011, and 0.015) nanofiber networks to tune sub-bandg...

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Effect of 0.01%, 0.02% and 0.04% Atropine eyedrops vs Placebo on myopia incidence in pre-myopic children: A randomized clinical trial

Xiangui He 2026-10-23 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Low-concentration atropine demonstrated a dose-response relationship for reducing cumulative myopia incidence and slowing refractive progression in pre-myopic children, but no such relationship on axial length. This preliminary evidence suggests potential for early myopia prevention, bu...

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Diagnosis of a chiasmal lesion compressing optic chiasm in patients with severe retinal degeneration: a two-case report

Kyung-Ah Park 2026-10-23 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Optic chiasmal compression should be considered in patients with severe retinal degeneration presenting with progressive visual decline that cannot be fully explained by retinal pathology. Visual field testing and timely neuroimaging, particularly MRI, are essential for accurate diagnos...

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https://pubmed.ncbi.nlm.nih.gov/42919189/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1xePBFBNvSIegfqCbvp45N3V9WgCNCS63Z1PLmhwJSPGd18QMT&fc=None&ff=20261024193342&v=2.20.1

Reading Arabic Under Simulated Low Vision: Orthographic Complexity Shapes the Effect of Contrast Polarity

👤 Mosaad Alhassan
📅 2026-10-23 📖 1 min 📝 45 words

LOW VISION

Summary: CONCLUSION: Arabic relies on small, high-spatial-frequency features such as dots for letter discrimination, which appear to modulate how contrast polarity interacts with optical degradation at threshold. These findings provide preliminary evidence that may guide future studies of reading performance...

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Screening-based prevalence and axis distribution of astigmatism among children and adolescents in Lintong District, Xi'an: a large-scale school-based cross-sectional study

👤 Zhaojiang Du
📅 2026-10-23 📖 1 min 📝 51 words

LOW VISION

Summary: CONCLUSIONS: Astigmatism was common in this large district school screening dataset and differed by school stage, axis type, and refractive-status category. The findings may help local screening programs plan re-examination and referral pathways, but they should not be interpreted as cycloplegic cli...

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Genome mining and comparative genome analysis of Pseudomonas fluorescens revealed diverse biosynthetic and metabolic potential

👤 Farida Begum 📅 2026-10-24 📖 1 min 📝 70 words LOW VISION

Summary: Bacterial genome sequencing has been used to identify a myriad of bioactive compounds that have yet to be characterised. Pseudomonas fluorescens is a widely distributed Gram-negative bacterium known for its plant growth-promoting traits and production of diverse secondary metabolites. P. fluorescens...

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Clinical profiles, referral patterns and low vision aid prescribing across four rehabilitation centres in Ghana: protocol for a multicentre retrospective record review

Werner Eisenbarth 2026-10-24 1 min 79 words LOW VISION

Summary: BACKGROUND: Low vision, defined as best-corrected visual acuity of less than 6/18 in the better eye or a visual field constriction of 20° or less, is a condition in which residual vision may be optimized through optical and non-optical aids. Despite the importance of low vision services in maximizin...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

Xin Zhou 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

👤 Peng Chen 📅 2026-10-23 📖 1 min 📝 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

👤 Xiaojun Zhou 📅 2026-10-23 📖 1 min 📝 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

👤 Hussein Youssef 📅 2026-10-23 📖 1 min 📝 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

👤 Leticia González 📅 2026-10-23 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Nonpharmacological Interventions in Degenerative Cerebellar Ataxia: A Systematic Review

👤 Pramod Kumar Pal 📅 2026-10-23 📖 1 min 📝 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: NPI may confer short-term symptomatic and functional benefits in degenerative cerebellar ataxia, particularly in axial motor domains, supported by the largest evidence for structured rehabilitation followed by neuromodulation, although heterogeneous. Future multicenter, biomarker-inform...

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Effects of Orbitofrontal Cortex tDCS on Cognitive Control in Individuals With Obesity: A Randomized Sham-Controlled Trial

👤 Stephanie L Borgland 📅 2026-10-23 📖 1 min 📝 43 words TDCS TACS TRNS

Summary: CONCLUSIONS: Unlike what has been reported in lean individuals, a single session of OFC-targeted tDCS is insufficient to modify cognitive control in individuals with obesity. However, the cognitive effects of a single session of OFC stimulation depend on individual neurocognitive characteristics in ...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

Xuqun You
2026-10-22 1 min 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

Rihui Li 2026-10-23 1 min 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

Georg S Kranz 2026-10-23 1 min 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

👤 Esra Suzen 📅 2026-10-23 📖 1 min 📝 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

👤 Lulu Song 📅 2026-10-24 📖 1 min 📝 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

Qingguo Wei 2026-10-23 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

👤 Jiayuan He 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Training Free Signal Enhancement by Random Sampling for Steady State Visual Evoked Potential BCI

👤 Yoshikazu Washizawa 📅 2026-10-23 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: The steady-state visual evoked potential (SSVEP) is widely used for
electroencephalography (EEG)
based brain computer interfaces (BCIs).
In order to determine the target commands for SSVEP-BCIs,
deep neural networks have been widely used.
However, learning-based approaches requir...

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Decoding attempted handwritten characters based on temporal endovascular neural features

Peter E Yoo 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: We evaluated temporal neural features for motor decoding using a minimally-invasive endovascular brain-computer interfaces (BCI). We hypothesised that relatively long compound movements generate distinct patterns of neural activity that are encoded by temporal features.

Approach. A single p...

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Transformer networks are robust learners of source localization for EEG measurements

Moritz Dannhauer 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

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Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

Chen Wang 2026-10-23 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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Generalizable neural decoding for clinically scalable brain-computer interfaces

👤 Jie Chen 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces promise restored communication and movement, yet accuracy achieved under development rarely survives new sessions, users, tasks, or hardware. Generalization, not peak performance, determines clinical scalability. Here we examine generalizable neural decoding from developmen...

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Polygenicity shapes distinct brain patterns and clinical profiles of bipolar disorder

👤 Shaohua Hu 📅 2026-10-23 📖 1 min 📝 41 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Our findings reveal associations between polygenicity and distinct BD brain patterns and clinical profiles. We identified ancestry-specific causal variants and bidirectional BD-IDP relationships. PRS associations with depressive symptoms, anxiety, suicide risk, and metabolic phenotypes ...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

📅 2026-07-30 📖 1 min 📝 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

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http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

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http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

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Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

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The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

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Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

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Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

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Bbtree: Fastest Ordered Map

👤 aoli-al 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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The Importance of Limitations for Creativity and Learning

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We should be more ambitious with the software we build

👤 ltratt 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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Argus – a free, open-source agentic security operations platform – MIT

👤 humxn 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

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http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

Adriel Carridice
2026-06-22 1 min 0 words

BRAIN

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Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

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https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

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Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

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Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

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Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

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Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

👤 Deidre Artis 📅 2026-07-17 📖 1 min 📝 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

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State-specific dynamic functional connectivity disturbances in spinal cord injury-related neuropathic pain

📖 1 min 📝 29 words BRAIN RESEARCH

> **Summary:** Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chuchu Sun, Haotian Xin, Ling Wang, Qian Chen, Yu Wang, Beining Yang, Xin Chen, Fang Li, Jie Lu, Nan Chen

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https://www.sciencedirect.com/science/article/pii/S0006899326004658?dgcid=rss_sd_all

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

> **Summary:** Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ultrabright Quasi-Compact Quantum Dot Assembly for Sensitive and Color-Fixed Immunoassays

👤 Liang Huang 📅 2026-10-23 📖 1 min 📝 63 words LOW VISION

Summary: Home-based immunoassay platforms constitute critical tools for the rapid screening of infectious respiratory viruses. Notably, viral loads can remain low at certain stages yet vary over several orders of magnitude during infection, and such variations are closely associated with the disease progress...

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Supercritical Carbon Dioxide-Peracetic Acid Treatment Inactivates Low-Level Bacterial Inocula in Collagen Analogue-Phosphorylcholine Hydrogels while Preserving Functional Properties

👤 May Griffith
📅 2026-10-23 📖 1 min 📝 57 words

LOW VISION

Summary: Terminal sterilization of hydrogels is challenging because their high-water-content, chemical structures, and cell-friendly properties are often altered by conventional sterilization methods. Hydrogel implants comprising collagen-like peptide (CLP), polyethylene glycol (PEG), and poly (2-methacryloy...

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A Bidirectional Photosensitive Multimodal PdSe2/MoS2 van der Waals Heterostructure Memristor Array for In-Sensor Neuromorphic Vision

👤 Jun He 📅 2026-10-23 📖 1 min 📝 60 words LOW VISION

Summary: Optoelectronic van der Waals (vdW) memristors integrating photosensing, memory, and computing capabilities are promising for artificial vision but are limited by single photoresponse modes, low yields, and high device variability. Here, we demonstrate a 1-kb PdSe2/MoS2 van der Waals heterostructure ...

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Al-Stoichiometry-Engineered Defect Profiles in InAlZnO Nanofibers for Balanced Charge Transport and Optical Plasticity

👤 Fengyun Wang
📅 2026-10-23 📖 1 min 📝 64 words

LOW VISION

Summary: Metal oxide optoelectronic synapses are promising for neuromorphic vision, but combining a low dark current with sustained photoinduced plasticity remains challenging. Here, Al stoichiometry is engineered in electrospun In1AlxZn0.08O (x = 0.007, 0.011, and 0.015) nanofiber networks to tune sub-bandg...

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Effect of 0.01%, 0.02% and 0.04% Atropine eyedrops vs Placebo on myopia incidence in pre-myopic children: A randomized clinical trial

Xiangui He 2026-10-23 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Low-concentration atropine demonstrated a dose-response relationship for reducing cumulative myopia incidence and slowing refractive progression in pre-myopic children, but no such relationship on axial length. This preliminary evidence suggests potential for early myopia prevention, bu...

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Diagnosis of a chiasmal lesion compressing optic chiasm in patients with severe retinal degeneration: a two-case report

Kyung-Ah Park 2026-10-23 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Optic chiasmal compression should be considered in patients with severe retinal degeneration presenting with progressive visual decline that cannot be fully explained by retinal pathology. Visual field testing and timely neuroimaging, particularly MRI, are essential for accurate diagnos...

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Reading Arabic Under Simulated Low Vision: Orthographic Complexity Shapes the Effect of Contrast Polarity

👤 Mosaad Alhassan
📅 2026-10-23 📖 1 min 📝 45 words

LOW VISION

Summary: CONCLUSION: Arabic relies on small, high-spatial-frequency features such as dots for letter discrimination, which appear to modulate how contrast polarity interacts with optical degradation at threshold. These findings provide preliminary evidence that may guide future studies of reading performance...

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Screening-based prevalence and axis distribution of astigmatism among children and adolescents in Lintong District, Xi'an: a large-scale school-based cross-sectional study

Zhaojiang Du
2026-10-23 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Astigmatism was common in this large district school screening dataset and differed by school stage, axis type, and refractive-status category. The findings may help local screening programs plan re-examination and referral pathways, but they should not be interpreted as cycloplegic cli...

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Genome mining and comparative genome analysis of Pseudomonas fluorescens revealed diverse biosynthetic and metabolic potential

👤 Farida Begum 📅 2026-10-24 📖 1 min 📝 70 words LOW VISION

Summary: Bacterial genome sequencing has been used to identify a myriad of bioactive compounds that have yet to be characterised. Pseudomonas fluorescens is a widely distributed Gram-negative bacterium known for its plant growth-promoting traits and production of diverse secondary metabolites. P. fluorescens...

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Clinical profiles, referral patterns and low vision aid prescribing across four rehabilitation centres in Ghana: protocol for a multicentre retrospective record review

Werner Eisenbarth 2026-10-24 1 min 79 words LOW VISION

Summary: BACKGROUND: Low vision, defined as best-corrected visual acuity of less than 6/18 in the better eye or a visual field constriction of 20° or less, is a condition in which residual vision may be optimized through optical and non-optical aids. Despite the importance of low vision services in maximizin...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

> **Summary:** CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

> **Summary:** Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

Xin Zhou 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

Peng Chen 2026-10-23 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

Xiaojun Zhou 2026-10-23 1 min 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

👤 Hussein Youssef 📅 2026-10-23 📖 1 min 📝 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

👤 Leticia González 📅 2026-10-23 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Nonpharmacological Interventions in Degenerative Cerebellar Ataxia: A Systematic Review

👤 Pramod Kumar Pal 📅 2026-10-23 📖 1 min 📝 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: NPI may confer short-term symptomatic and functional benefits in degenerative cerebellar ataxia, particularly in axial motor domains, supported by the largest evidence for structured rehabilitation followed by neuromodulation, although heterogeneous. Future multicenter, biomarker-inform...

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Effects of Orbitofrontal Cortex tDCS on Cognitive Control in Individuals With Obesity: A Randomized Sham-Controlled Trial

👤 Stephanie L Borgland 📅 2026-10-23 📖 1 min 📝 43 words TDCS TACS TRNS

Summary: CONCLUSIONS: Unlike what has been reported in lean individuals, a single session of OFC-targeted tDCS is insufficient to modify cognitive control in individuals with obesity. However, the cognitive effects of a single session of OFC stimulation depend on individual neurocognitive characteristics in ...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

Xuqun You
2026-10-22 1 min 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

👤 Rihui Li 📅 2026-10-23 📖 1 min 📝 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

Lulu Song 2026-10-24 1 min 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

Qingguo Wei 2026-10-23 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

👤 Jiayuan He 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Training Free Signal Enhancement by Random Sampling for Steady State Visual Evoked Potential BCI

👤 Yoshikazu Washizawa 📅 2026-10-23 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: The steady-state visual evoked potential (SSVEP) is widely used for
electroencephalography (EEG)
based brain computer interfaces (BCIs).
In order to determine the target commands for SSVEP-BCIs,
deep neural networks have been widely used.
However, learning-based approaches requir...

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Decoding attempted handwritten characters based on temporal endovascular neural features

Peter E Yoo 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: We evaluated temporal neural features for motor decoding using a minimally-invasive endovascular brain-computer interfaces (BCI). We hypothesised that relatively long compound movements generate distinct patterns of neural activity that are encoded by temporal features.

Approach. A single p...

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Transformer networks are robust learners of source localization for EEG measurements

Moritz Dannhauer 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

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Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

Chen Wang 2026-10-23 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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Generalizable neural decoding for clinically scalable brain-computer interfaces

👤 Jie Chen 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces promise restored communication and movement, yet accuracy achieved under development rarely survives new sessions, users, tasks, or hardware. Generalization, not peak performance, determines clinical scalability. Here we examine generalizable neural decoding from developmen...

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Polygenicity shapes distinct brain patterns and clinical profiles of bipolar disorder

👤 Shaohua Hu 📅 2026-10-23 📖 1 min 📝 41 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Our findings reveal associations between polygenicity and distinct BD brain patterns and clinical profiles. We identified ancestry-specific causal variants and bidirectional BD-IDP relationships. PRS associations with depressive symptoms, anxiety, suicide risk, and metabolic phenotypes ...

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Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Keep Android Open 🔓

📅 2026-02-26 📖 3 min 📝 625 words FMHY

Summary: <h3 id="android-will-become-a-locked-down-platform-in-less-than-200-days" tabindex="-1">Android will become a locked-down platform in less than 200 days. </h3> <p>In August 2025, Google announced th...

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https://fmhy.net/posts/KeepAndroidOpen

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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https://fmhy.net/posts/april-2026

Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1097 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

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https://fmhy.net/posts/oct-2026

LiteLLM’s root-cause report: the key behind the Oct 21 release attack was leaked on May 27 and never

👤 /u/correctMeIfImcorrect 📅 2026-10-24 📖 1 min 📝 140 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> https://youtu.be/vxE3XbcLCa8 LiteLLM published its root-cause report on Oct 23. According to LiteLLM, a master key for its Agent Platform was committed publicly on May 27. The team found the leak and rewrote the com...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xehnro/litellms_rootcause_report_the_key_behind_the_oct/

Why is AI generated language so hard to understand?

👤 canjobear 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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Dream-1

👤 GaggiX 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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Danish Government silently drops plan to dump MS Office

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PHP Foundation 'under-funded' with one full-time staff member, director says

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John Carmack on the Goose Language: multiple data stacks and relative references

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The Goose Programming Language

👤 andsoitis 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://aardappel.github.io/goose/ Comments URL: https://news.ycombinator.com/item?id=50224300 Points: 87 # Comments: 33

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No Signs of AI Payoff Outside the Tech Sector

👤 the-mitr 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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Steal My Bitcoin

👤 cloudlandsdev 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://stealmybitcoin.com/ Comments URL: https://news.ycombinator.com/item?id=50224916 Points: 27 # Comments: 1

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There are 13 ways to analyse a replication study, but only one of them is coherent.

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-05-03 📖 6 min 📝 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3483 words

TWENTY PERCENT STATISTICIAN

Summary: <p>The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Science Framework. After February 19 all projects will become read-only, and eventually, they anticipate that all private projects will be deleted. The OSF has raised a generation of ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice
2026-09-23 1 min 0 words

BRAIN

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https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

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https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

👤 Deidre Artis 📅 2026-09-11 📖 1 min 📝 14 words EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

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https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

An adaptive transformer-based model with entropy-driven knowledge distillation for single-channel EEG REM sleep detection

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-62389-3

CLSP-REQA: a real-time quality-aware closed-loop seizure prediction framework with Mamba-BiLSTM and confidence-gated intervention

👤 Quansheng Ren 📅 2026-09-30 📖 1 min 📝 304 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionReliable seizure prediction is a prerequisite for closed-loop neurostimulation therapy, yet existing methods rarely account for the variability in EEG signal quality encountered in real-world deployment, and the overwhelming majority adopt non-strict evaluation protocols that overestimat...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1888743

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong 2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

HQML-ICU: hybrid quantum-classical machine learning for ICU mortality prediction using MIMIC-IV electronic health records

Mohammed Farsi 2026-10-09 1 min 277 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Accurate prediction of mortality in the intensive care unit (ICU) is crucial for informing clinical decision-making and optimizing resource allocation. Although machine learning methodologies have enhanced traditional severity-of-illness scoring systems, they encounter intrinsic limitations in model...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1900553

Adaptive neuro-fuzzy risk index for safe and interpretable lane-change decisions in Level-3 autonomous driving

Jianwei Li 2026-10-09 1 min 250 words FRONTIERS NEUROROBOTICS

Summary: IntroductionHighway lane-change maneuvers pose a safety challenge for SAE Level 3 autonomous driving because conventional threshold-based decisions and isolated surrogate safety measures may inadequately capture the coupling between longitudinal braking demand and evolving inter-vehicle gaps. This s...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1918469

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ultrabright Quasi-Compact Quantum Dot Assembly for Sensitive and Color-Fixed Immunoassays

👤 Liang Huang 📅 2026-10-23 📖 1 min 📝 63 words LOW VISION

Summary: Home-based immunoassay platforms constitute critical tools for the rapid screening of infectious respiratory viruses. Notably, viral loads can remain low at certain stages yet vary over several orders of magnitude during infection, and such variations are closely associated with the disease progress...

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Supercritical Carbon Dioxide-Peracetic Acid Treatment Inactivates Low-Level Bacterial Inocula in Collagen Analogue-Phosphorylcholine Hydrogels while Preserving Functional Properties

👤 May Griffith
📅 2026-10-23 📖 1 min 📝 57 words LOW VISION

Summary: Terminal sterilization of hydrogels is challenging because their high-water-content, chemical structures, and cell-friendly properties are often altered by conventional sterilization methods. Hydrogel implants comprising collagen-like peptide (CLP), polyethylene glycol (PEG), and poly (2-methacryloy...

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A Bidirectional Photosensitive Multimodal PdSe2/MoS2 van der Waals Heterostructure Memristor Array for In-Sensor Neuromorphic Vision

👤 Jun He 📅 2026-10-23 📖 1 min 📝 60 words LOW VISION

Summary: Optoelectronic van der Waals (vdW) memristors integrating photosensing, memory, and computing capabilities are promising for artificial vision but are limited by single photoresponse modes, low yields, and high device variability. Here, we demonstrate a 1-kb PdSe2/MoS2 van der Waals heterostructure ...

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Al-Stoichiometry-Engineered Defect Profiles in InAlZnO Nanofibers for Balanced Charge Transport and Optical Plasticity

👤 Fengyun Wang
📅 2026-10-23 📖 1 min 📝 64 words

LOW VISION

Summary: Metal oxide optoelectronic synapses are promising for neuromorphic vision, but combining a low dark current with sustained photoinduced plasticity remains challenging. Here, Al stoichiometry is engineered in electrospun In1AlxZn0.08O (x = 0.007, 0.011, and 0.015) nanofiber networks to tune sub-bandg...

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Effect of 0.01%, 0.02% and 0.04% Atropine eyedrops vs Placebo on myopia incidence in pre-myopic children: A randomized clinical trial

Xiangui He 2026-10-23 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Low-concentration atropine demonstrated a dose-response relationship for reducing cumulative myopia incidence and slowing refractive progression in pre-myopic children, but no such relationship on axial length. This preliminary evidence suggests potential for early myopia prevention, bu...

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Diagnosis of a chiasmal lesion compressing optic chiasm in patients with severe retinal degeneration: a two-case report

Kyung-Ah Park 2026-10-23 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Optic chiasmal compression should be considered in patients with severe retinal degeneration presenting with progressive visual decline that cannot be fully explained by retinal pathology. Visual field testing and timely neuroimaging, particularly MRI, are essential for accurate diagnos...

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Reading Arabic Under Simulated Low Vision: Orthographic Complexity Shapes the Effect of Contrast Polarity

👤 Mosaad Alhassan
📅 2026-10-23 📖 1 min 📝 45 words

LOW VISION

Summary: CONCLUSION: Arabic relies on small, high-spatial-frequency features such as dots for letter discrimination, which appear to modulate how contrast polarity interacts with optical degradation at threshold. These findings provide preliminary evidence that may guide future studies of reading performance...

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Screening-based prevalence and axis distribution of astigmatism among children and adolescents in Lintong District, Xi'an: a large-scale school-based cross-sectional study

Zhaojiang Du
2026-10-23 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Astigmatism was common in this large district school screening dataset and differed by school stage, axis type, and refractive-status category. The findings may help local screening programs plan re-examination and referral pathways, but they should not be interpreted as cycloplegic cli...

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Genome mining and comparative genome analysis of Pseudomonas fluorescens revealed diverse biosynthetic and metabolic potential

👤 Farida Begum 📅 2026-10-24 📖 1 min 📝 70 words LOW VISION

Summary: Bacterial genome sequencing has been used to identify a myriad of bioactive compounds that have yet to be characterised. Pseudomonas fluorescens is a widely distributed Gram-negative bacterium known for its plant growth-promoting traits and production of diverse secondary metabolites. P. fluorescens...

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Clinical profiles, referral patterns and low vision aid prescribing across four rehabilitation centres in Ghana: protocol for a multicentre retrospective record review

👤 Werner Eisenbarth 📅 2026-10-24 📖 1 min 📝 79 words LOW VISION

Summary: BACKGROUND: Low vision, defined as best-corrected visual acuity of less than 6/18 in the better eye or a visual field constriction of 20° or less, is a condition in which residual vision may be optimized through optical and non-optical aids. Despite the importance of low vision services in maximizin...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

👤 Martina M Mensi 📅 2026-06-25 📖 1 min 📝 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

Xin Zhou 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

Peng Chen 2026-10-23 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

Xiaojun Zhou 2026-10-23 1 min 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

👤 Hussein Youssef 📅 2026-10-23 📖 1 min 📝 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

👤 Leticia González 📅 2026-10-23 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Nonpharmacological Interventions in Degenerative Cerebellar Ataxia: A Systematic Review

👤 Pramod Kumar Pal 📅 2026-10-23 📖 1 min 📝 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: NPI may confer short-term symptomatic and functional benefits in degenerative cerebellar ataxia, particularly in axial motor domains, supported by the largest evidence for structured rehabilitation followed by neuromodulation, although heterogeneous. Future multicenter, biomarker-inform...

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Effects of Orbitofrontal Cortex tDCS on Cognitive Control in Individuals With Obesity: A Randomized Sham-Controlled Trial

👤 Stephanie L Borgland 📅 2026-10-23 📖 1 min 📝 43 words TDCS TACS TRNS

Summary: CONCLUSIONS: Unlike what has been reported in lean individuals, a single session of OFC-targeted tDCS is insufficient to modify cognitive control in individuals with obesity. However, the cognitive effects of a single session of OFC stimulation depend on individual neurocognitive characteristics in ...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

👤 Rihui Li 📅 2026-10-23 📖 1 min 📝 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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https://pubmed.ncbi.nlm.nih.gov/42915317/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261024142004&v=2.20.1

The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

Lulu Song 2026-10-24 1 min 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

Qingguo Wei 2026-10-23 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

👤 Jiayuan He 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Training Free Signal Enhancement by Random Sampling for Steady State Visual Evoked Potential BCI

👤 Yoshikazu Washizawa 📅 2026-10-23 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: The steady-state visual evoked potential (SSVEP) is widely used for
electroencephalography (EEG)
based brain computer interfaces (BCIs).
In order to determine the target commands for SSVEP-BCIs,
deep neural networks have been widely used.
However, learning-based approaches requir...

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Decoding attempted handwritten characters based on temporal endovascular neural features

Peter E Yoo 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: We evaluated temporal neural features for motor decoding using a minimally-invasive endovascular brain-computer interfaces (BCI). We hypothesised that relatively long compound movements generate distinct patterns of neural activity that are encoded by temporal features.

Approach. A single p...

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Transformer networks are robust learners of source localization for EEG measurements

Moritz Dannhauer 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

🔗 **Read full article:**
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Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

Chen Wang 2026-10-23 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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https://pubmed.ncbi.nlm.nih.gov/42915561/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261024142001&v=2.20.1

Generalizable neural decoding for clinically scalable brain-computer interfaces

👤 Jie Chen 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

> **Summary:** Brain-computer interfaces promise restored communication and movement, yet accuracy achieved under development rarely survives new sessions, users, tasks, or hardware. Generalization, not peak performance, determines clinical scalability. Here we examine generalizable neural decoding from developmen...

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Polygenicity shapes distinct brain patterns and clinical profiles of bipolar disorder

👤 Shaohua Hu 📅 2026-10-23 📖 1 min 📝 41 words BRAIN COMPUTER INTERFACE

> **Summary:** CONCLUSIONS: Our findings reveal associations between polygenicity and distinct BD brain patterns and clinical profiles. We identified ancestry-specific causal variants and bidirectional BD-IDP relationships. PRS associations with depressive symptoms, anxiety, suicide risk, and metabolic phenotypes ...

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Psychomusicology: A resounding closing cadence.

📅 2024-01-22 📖 1 min 📝 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual's cognition in solving a task. However, very little is known about how an individual's conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000400

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/may-2026

Root shell on Samsung TV

👤 testing22321 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://twitter.com/log4j/status/2113014142600953868?s=46 Comments URL: https://news.ycombinator.com/item?id=50218633 Points: 440 #...

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FreeDNS

👤 esher 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://freedns.afraid.org/ Comments URL: https://news.ycombinator.com/item?id=50222133 Points: 151 # Comments: 36

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https://freedns.afraid.org/

Oai-math-revised: a community hub for revising OpenAI's math write-ups

👤 tokepi 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://terrytao.wordpress.com/2026/10/24/oai-math-revised-a-community-hub-for-revising-openais-math-write-ups/ Comments URL: <a href="https://news.yco...

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Neural chip decodes interactions between brain regions

👤 giuliomagnifico 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://actu.epfl.ch/news/neural-chip-decodes-interactions-between-brain-r-2/ Comments URL: <a href="https://news.ycombinator.com/item?id=50222525">https://news.ycombinator.com/item?id=...

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https://actu.epfl.ch/news/neural-chip-decodes-interactions-between-brain-r-2/

Roku developer repositories moved from GitHub to GitLab

👤 cxr 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.roku.com/developer/gitlab-move Comments URL: https://news.ycombinator.com/item?id=50222622 Points: 121 # Comments: 69

🔗 **Read full article:**
https://blog.roku.com/developer/gitlab-move

The St. Helena 5 pence coin shows Jonathan, a roughly 190 year old tortoise

👤 schleck8 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://de.numista.com/9866 Comments URL: https://news.ycombinator.com/item?id=50222927 Points: 38 # Comments: 3

🔗 **Read full article:**
https://de.numista.com/9866

Are meta-scientists ignoring philosophy of science?

noreply@blogger.com (Daniel Lakens)
2025-07-04 8 min 1681 words

TWENTY PERCENT STATISTICIAN

Summary: Are meta-scientists ignoring philosophy of science (PoS)? Are they re-inventing the wheel? A recent panel at the Metascience conference engaged with this question, and the first sentence of the abstract states “Critics argue t...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/are-meta-scientists-ignoring-philosophy.html

Type S and M errors as a “rhetorical tool”

noreply@blogger.com (Daniel Lakens)
2025-09-28 17 min 3572 words

TWENTY PERCENT STATISTICIAN

Summary: Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

noreply@blogger.com (Daniel Lakens) 2025-10-29 27 min 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

Steve Luck 2023-06-23 2 min 420 words

ERP BOOT CAMP

Summary: You can access the recording h.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

Summary: The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

Deidre Artis 2026-08-18 1 min 14 words EMBS

Summary: The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Alterations in the populations of peripheral immune cells in the blood of a mouse model of progressive myoclonus epilepsy of Lafora type

1 min 17 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): L. Baños-Carrión, A. Campos-Rodriguez, M.A. Garcia-Gimeno, P. Sanz

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004580?dgcid=rss_sd_all

Addressing the between-group comparison problem: Detecting differences between correlation matrix populations due to single-variable perturbations for resting state fMRI

📖 1 min 📝 16 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Itamar Faran, Michael Peer, Shahar Arzy, Yuval Benjamini

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926005549?dgcid=rss_sd_all

A comparison of continuous-wave fNIRS with quantitative fMRI-derived indices of brain function in the visual cortex

📖 1 min 📝 42 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Giulia Rocco, Lucie Chalet, Elizabeth J. Fear, Sara Pomante, Francesca Graziano, Davide Di Censo, Manuela Carriero, Édouard Delaire, Fabrizio Esposito, Mauro Gianni Perrucci, Cosimo Del Gratta, David P...

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006051?dgcid=rss_sd_all

Circadian and sleep-wake modulation of EEG functional connectivity across brain states and Its associations with cognition in humans

📖 1 min 📝 24 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Alpar S Lazar, Zsolt I Lazar, Nayantara Santhi, June C Lo, John A Groeger, Derk-Jan Dijk

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006166?dgcid=rss_sd_all

Post-Learning Reactivation of Memories for Naturalistic Continuous Experiences

👤 Wolf, N., Antony, J. W., Delarazan, A. I., Reagh, Z. M., Michelmann, S., Ranganath, C., Liu, X. L.

📅 2026-10-24 📖 1 min 📝 148 words BIORXIV NEUROSCIENCE

> **Summary:** Memories are not static records of the past; rather, the act of remembering can change how an event is recalled. Intriguingly, several studies also suggest that memories may be reactivated during rest. However, reactivation has largely been studied for spatial tasks and discrete stimuli, and we know...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.20.760554v1?rss=1

The Impact of Noise on Memorability in Minds and Machines

Ding, H., Wiener, M. 2026-10-24 1 min 236 words BIORXIV NEUROSCIENCE

Summary: Some images are consistently remembered better than others, a stimulus property known as memorability, and deep neural networks (DNNs) are increasingly used to estimate it in place of human observers. A recent account proposes that memorable images are processed more efficiently. We argue that memor...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.19.759153v1?rss=1

Microglia support regeneration of rod photoreceptors in larval zebrafish retina through Nrf2-independent mechanisms

Morales, M., Mitchell, D. M. 2026-10-24 1 min 187 words BIORXIV NEUROSCIENCE

Summary: The molecular and cellular components of successful regeneration of complex tissues in zebrafish are a major area of investigation. Previous work by our lab indicates a redox function for retinal microglia in the first 48 hours following acute injury to rod photoreceptors in the larval zebrafish ret...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761077v1?rss=1

Human cortical slices reveal pulse-number-dependent synaptic plasticity during theta-burst magnetic stimulation

Galanis, C., Straehle, J., Turi, Z., Beck, J., Vlachos, A.

2026-10-24 1 min 169 words

BIORXIV NEUROSCIENCE

Summary: Theta-burst transcranial magnetic stimulation is widely used to modulate human brain activity in clinical and experimental settings, yet how clinically relevant stimulation protocols engage synaptic plasticity in human cortical circuits remains poorly understood. Here, we combined controlled magneti...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761311v1?rss=1

The role of contextual similarity on brain activity patterns for associated events

Gurguryan, L., Olivier, J. M., Dimsdale-Zucker, H. R.

2026-10-24 1 min 228 words

BIORXIV NEUROSCIENCE

Summary: Our everyday lives are characterized by repeated contexts, such as returning to work in the same office every day. Although the physical office may be identical, other features vary--e.g., time of day, light conditions, one's internal state--leading to differing degrees of similarity, even among put...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.19.760268v1?rss=1

`Silent' mitral cells emerge from common feature subtraction

Tootoonian, S., Kollo, M., Schaefer, A. T.

2026-10-24 1 min 199 words

BIORXIV NEUROSCIENCE

Summary: Whole-cell recordings from olfactory bulb mitral/tufted cells in awake mice revealed an inverse relationship between baseline activation and odour responses wherein cells that were spontaneously active tended to be inhibited by odours, while `silent' cells, quiet at baseline, tended to have vigorous...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.761107v1?rss=1

From hours to days: prolonging the functional viability of acute brain slices to >36 h

Vylekzhanina, E., Habelt, L., Creamer, A., Cameron de Abos y Padilla, C., Diester, I., Niessing, M., Hochstrate, P., Coulon, P.

2026-10-24 1 min 139 words

BIORXIV NEUROSCIENCE

Summary: We have developed an experimental protocol for acute brain slices of juvenile and adult mice which, in our hands, enables whole-cell patch-clamp experiments on healthy neurons with intact electrophysiological characteristics for more than 36 hours after preparation. This drastically expands the conv...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.760385v1?rss=1

Transient noradrenergic signals act as online controllers of perceptual computation

👤 Ghosh, S., Maunsell, J. H.
📅 2026-10-24 📖 1 min 📝 237 words BIORXIV NEUROSCIENCE

Summary: Neuromodulatory signals influence perception, arousal and learning, but their effects on behavioral performance can arise from fundamentally different computations. A transient signal could alter a persistent operating state, control an ongoing computation, or provide a teaching signal that modifies...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.760810v1?rss=1

Reduced-observable dynamic mode decomposition for identifying theta and beta activity in working-memory EEG

👤 Chavez-Huerta, D., Khosravi, M.
📅 2026-10-24 📖 1 min 📝 199 words BIORXIV NEUROSCIENCE

Summary: The analysis of transient oscillatory activity in neural recordings remains challenging for linear methods, particularly when signals are nonlinear, noisy, and distributed across multiple channels. This paper introduces a Reduced-Observable Extended Dynamic Mode Decomposition (RO-eDMD) algorithm usi...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.20.760227v1?rss=1

Glutamatergic Neurons in caudal Pontine Reticular Nucleus (PRNc) Mediate Backward Locomotion in mice

👤 Seo, M. B., Huang, J. J., Zhang, G.-W., Voong, K., Tao, H. W., Zhang, L. I.

📅 2026-10-24 📖 1 min

📝 182 words

BIORXIV NEUROSCIENCE

Summary: Locomotion is essential for animals to adaptively navigate their environment. However, despite extensive investigation into neural mechanisms underlying forward movement, the brain circuits controlling backward locomotion, a critical motor behavior for navigating confined spaces and avoiding threats...

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Human brain organoids reveal MGE interneuron defects independent of mTOR-hyperactivation in genetic focal epilepsy

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NATURE NEUROSCIENCE SUBJECTS

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Reliability-based arbitration reveals heterogeneous model-based and model-free reinforcement learning strategies linked to compulsivity and impulsivity

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

👤 Jianfeng Hu 📅 2026-10-16 📖 1 min 📝 222 words FRONTIERS NEUROROBOTICS

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1925089

Mapping unintended trigeminal nerve activation during transcranial alternating current stimulation (tACS): a computational study

👤 Sulagna Sahu, Munish Chauhan, Saurav Z K Sajib, Enock Boakye and Rosalind J Sadleir

📅 2026-10-15 📖 1 min 📝 211 words JOURNAL NEURAL ENGINEERING

Summary: Objective. To systematically characterize activation thresholds for peripheral trigeminal nerve branches under clinical transcranial alternating current stimulation (tACS) parameters using a computational modeling approach. Approach. An MRI-derived finite element head model was coupled with multicom...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb074

Involuntary facial muscle activity during imagined vocalisation contaminates EEG and enables emotion decoding

Yichen Tang, Paul M Corballis and Luke E Hallum

2026-10-15 1 min 281 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Decoding imagined speech from electroencephalography (EEG) recordings is potentially useful for brain-computer interfaces. Previous studies have focused on decoding semantic information from EEG, leaving the decoding of emotion — an important component of human communication — largely une...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb073

Assessing consciousness by near-nerve electrostimulation in intensive rehabilitation

Piergiuseppe Liuzzi, Sara Ballanti, Andrea Mannini, Bahia Hakiki, Rachele Burali, Tanita Toci, Maria Chiara Carrozza, Calogero Maria Oddo and Antonello Grippo

2026-10-15 1 min 172 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Consciousness diagnosis in prolonged disorders of consciousness (pDoC) relies on the Coma Recovery Scale–Revised, yet consciousness or its potential for recovery may escape behavioural detection. We tested whether near-nerve stimulation (NNS) of the median nerve, compared to standard surf...

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Prioritized nonlinear modeling of shared dynamics across neural populations

Trisha Jha, Dong Song and Maryam M Shanechi

2026-10-15 1 min 284 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Advances in neural recording technology enable simultaneous measurements across multiple brain regions, providing new opportunities to study inter-regional interactions. However, several challenges arise when developing nonlinear dynamical models of cross-region interactions. In particula...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

> **Summary:** The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ultrabright Quasi-Compact Quantum Dot Assembly for Sensitive and Color-Fixed Immunoassays

👤 Liang Huang 📅 2026-10-23 📖 1 min 📝 63 words LOW VISION

Summary: Home-based immunoassay platforms constitute critical tools for the rapid screening of infectious respiratory viruses. Notably, viral loads can remain low at certain stages yet vary over several orders of magnitude during infection, and such variations are closely associated with the disease progress...

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Supercritical Carbon Dioxide-Peracetic Acid Treatment Inactivates Low-Level Bacterial Inocula in Collagen Analogue-Phosphorylcholine Hydrogels while Preserving Functional Properties

👤 May Griffith
📅 2026-10-23 📖 1 min 📝 57 words

LOW VISION

Summary: Terminal sterilization of hydrogels is challenging because their high-water-content, chemical structures, and cell-friendly properties are often altered by conventional sterilization methods. Hydrogel implants comprising collagen-like peptide (CLP), polyethylene glycol (PEG), and poly (2-methacryloy...

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A Bidirectional Photosensitive Multimodal PdSe2/MoS2 van der Waals Heterostructure Memristor Array for In-Sensor Neuromorphic Vision

👤 Jun He 📅 2026-10-23 📖 1 min 📝 60 words LOW VISION

Summary: Optoelectronic van der Waals (vdW) memristors integrating photosensing, memory, and computing capabilities are promising for artificial vision but are limited by single photoresponse modes, low yields, and high device variability. Here, we demonstrate a 1-kb PdSe2/MoS2 van der Waals heterostructure ...

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Al-Stoichiometry-Engineered Defect Profiles in InAlZnO Nanofibers for Balanced Charge Transport and Optical Plasticity

👤 Fengyun Wang
📅 2026-10-23 📖 1 min 📝 64 words

LOW VISION

Summary: Metal oxide optoelectronic synapses are promising for neuromorphic vision, but combining a low dark current with sustained photoinduced plasticity remains challenging. Here, Al stoichiometry is engineered in electrospun In1AlxZn0.08O (x = 0.007, 0.011, and 0.015) nanofiber networks to tune sub-bandg...

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Effect of 0.01%, 0.02% and 0.04% Atropine eyedrops vs Placebo on myopia incidence in pre-myopic children: A randomized clinical trial

Xiangui He 2026-10-23 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Low-concentration atropine demonstrated a dose-response relationship for reducing cumulative myopia incidence and slowing refractive progression in pre-myopic children, but no such relationship on axial length. This preliminary evidence suggests potential for early myopia prevention, bu...

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Diagnosis of a chiasmal lesion compressing optic chiasm in patients with severe retinal degeneration: a two-case report

Kyung-Ah Park 2026-10-23 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Optic chiasmal compression should be considered in patients with severe retinal degeneration presenting with progressive visual decline that cannot be fully explained by retinal pathology. Visual field testing and timely neuroimaging, particularly MRI, are essential for accurate diagnos...

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Reading Arabic Under Simulated Low Vision: Orthographic Complexity Shapes the Effect of Contrast Polarity

👤 Mosaad Alhassan 📅 2026-10-23 📖 1 min 📝 45 words LOW VISION

Summary: CONCLUSION: Arabic relies on small, high-spatial-frequency features such as dots for letter discrimination, which appear to modulate how contrast polarity interacts with optical degradation at threshold. These findings provide preliminary evidence that may guide future studies of reading performance...

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Screening-based prevalence and axis distribution of astigmatism among children and adolescents in Lintong District, Xi'an: a large-scale school-based cross-sectional study

Zhaojiang Du
2026-10-23 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Astigmatism was common in this large district school screening dataset and differed by school stage, axis type, and refractive-status category. The findings may help local screening programs plan re-examination and referral pathways, but they should not be interpreted as cycloplegic cli...

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Genome mining and comparative genome analysis of Pseudomonas fluorescens revealed diverse biosynthetic and metabolic potential

👤 Farida Begum 📅 2026-10-24 📖 1 min 📝 70 words LOW VISION

Summary: Bacterial genome sequencing has been used to identify a myriad of bioactive compounds that have yet to be characterised. Pseudomonas fluorescens is a widely distributed Gram-negative bacterium known for its plant growth-promoting traits and production of diverse secondary metabolites. P. fluorescens...

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Clinical profiles, referral patterns and low vision aid prescribing across four rehabilitation centres in Ghana: protocol for a multicentre retrospective record review

Werner Eisenbarth 2026-10-24 1 min 79 words LOW VISION

Summary: BACKGROUND: Low vision, defined as best-corrected visual acuity of less than 6/18 in the better eye or a visual field constriction of 20° or less, is a condition in which residual vision may be optimized through optical and non-optical aids. Despite the importance of low vision services in maximizin...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

Xin Zhou 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

👤 Peng Chen 📅 2026-10-23 📖 1 min 📝 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

👤 Xiaojun Zhou 📅 2026-10-23 📖 1 min 📝 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

👤 Hussein Youssef 📅 2026-10-23 📖 1 min 📝 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

👤 Leticia González 📅 2026-10-23 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Nonpharmacological Interventions in Degenerative Cerebellar Ataxia: A Systematic Review

Pramod Kumar Pal 2026-10-23 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: NPI may confer short-term symptomatic and functional benefits in degenerative cerebellar ataxia, particularly in axial motor domains, supported by the largest evidence for structured rehabilitation followed by neuromodulation, although heterogeneous. Future multicenter, biomarker-inform...

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Effects of Orbitofrontal Cortex tDCS on Cognitive Control in Individuals With Obesity: A Randomized Sham-Controlled Trial

Stephanie L Borgland 2026-10-23 1 min 43 words TDCS TACS TRNS

Summary: CONCLUSIONS: Unlike what has been reported in lean individuals, a single session of OFC-targeted tDCS is insufficient to modify cognitive control in individuals with obesity. However, the cognitive effects of a single session of OFC stimulation depend on individual neurocognitive characteristics in ...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

👤 Rihui Li 📅 2026-10-23 📖 1 min 📝 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

👤 Esra Suzen 📅 2026-10-23 📖 1 min 📝 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

👤 Lulu Song 📅 2026-10-24 📖 1 min 📝 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

Qingguo Wei 2026-10-23 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

👤 Jiayuan He 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Training Free Signal Enhancement by Random Sampling for Steady State Visual Evoked Potential BCI

👤 Yoshikazu Washizawa 📅 2026-10-23 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: The steady-state visual evoked potential (SSVEP) is widely used for
electroencephalography (EEG)
based brain computer interfaces (BCIs).
In order to determine the target commands for SSVEP-BCIs,
deep neural networks have been widely used.
However, learning-based approaches requir...

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Decoding attempted handwritten characters based on temporal endovascular neural features

Peter E Yoo 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: We evaluated temporal neural features for motor decoding using a minimally-invasive endovascular brain-computer interfaces (BCI). We hypothesised that relatively long compound movements generate distinct patterns of neural activity that are encoded by temporal features.

Approach. A single p...

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Transformer networks are robust learners of source localization for EEG measurements

Moritz Dannhauer 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42914707/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261024113045&v=2.20.1

Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

Chen Wang 2026-10-23 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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https://pubmed.ncbi.nlm.nih.gov/42915561/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261024113045&v=2.20.1

Generalizable neural decoding for clinically scalable brain-computer interfaces

👤 Jie Chen 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces promise restored communication and movement, yet accuracy achieved under development rarely survives new sessions, users, tasks, or hardware. Generalization, not peak performance, determines clinical scalability. Here we examine generalizable neural decoding from developmen...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42919847/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261024113045&v=2.20.1

Polygenicity shapes distinct brain patterns and clinical profiles of bipolar disorder

👤 Shaohua Hu 📅 2026-10-23 📖 1 min 📝 41 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Our findings reveal associations between polygenicity and distinct BD brain patterns and clinical profiles. We identified ancestry-specific causal variants and bidirectional BD-IDP relationships. PRS associations with depressive symptoms, anxiety, suicide risk, and metabolic phenotypes ...

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Anyone Learning FastAPI? Let's Connect! 🚀

👤 /u/aryansking 📅 2026-10-24 📖 1 min 📝 59 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Hey everyone! I'm looking to connect with people who are learning FastAPI and interested in growing together. We can share knowledge, build real-world projects, and learn production-level backend development together. If you're learning FastAPI ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xe8o2s/anyone_learning_fastapi_lets_connect/

Resource Sharing: Memory Graph Web Debugger, visualize program state

👤 /u/Sea-Ad7805 📅 2026-10-24 📖 1 min 📝 154 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Programming courses often focus heavily on understanding code, while paying far less attention to understanding the program state. I've created **memory_graph**: a tool that visualizes the state of a Python program as it changes, step by step. It can help ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xe91v7/resource_sharing_memory_graph_web_debugger/

GPU-unlock: FP/BF16 instruction issue rate on Nvidia 3000's cards

👤 fragebogen 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/Zagitta/gpu-unlock Comments URL: https://news.ycombinator.com/item?id=50219700 Points: 24 # Comments: 15

🔗 **Read full article:**
https://github.com/Zagitta/gpu-unlock

Show HN: Autonomous root for Tcl Android TV via CVE-2022-22706

👤 t20n 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://gitlab.com/c4psu13/mali-pagecache-root Comments URL: https://news.ycombinator.com/item?id=50220042 Points: 21 # Comments: 2

🔗 **Read full article:**
https://gitlab.com/c4psu13/mali-pagecache-root

Google illegally cut down 250K trees in Finland for data center

👤 rustybolt 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nypost.com/2026/10/24/business/google-illegally-tore-down-250k-trees-for-finland-data-center-infuriating-locals-report/ Comment...

🔗 **Read full article:**
https://nypost.com/2026/10/24/business/google-illegally-tore-down-250k-trees-for-finland-data-center-infuriating-locals-report/

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

👤 Deidre Artis 📅 2026-09-29 📖 1 min 📝 14 words EMBS

> **Summary:** The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Vitamin B6 augments antidepressant activity of venlafaxine by restoring cortical EEG oscillations, behavior and oxidative biomarkers in chronic unpredictable mild stress (CUMS)-induced model of depression-like behavior

📖 1 min 📝 30 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Fashwa Khan Tareen, Sana Javaid, Waseem Ashraf, Syed Muhammad Muneeb Anjum, Rana Muhammad Zahid Mushtaq, Asad Abrar, Faleh Alqahtani, Imran Imran

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004506?dgcid=rss_sd_all

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

📖 1 min 📝 34 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004634?dgcid=rss_sd_all

Neurodegenerative diseases in Sub-Saharan Africa: diagnostic invisibility, biological heterogeneity and a precision-neuroscience agenda

📖 1 min 📝 11 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Maulilio John Kipanyula

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006962?dgcid=rss_sd_all

Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zuhal Abasiyanik, Lousin Moumdjian, Natalia Skorupa, Daphne Kos, Bart Van Wijmeersch, Miguel D’Haeseleer, Inge Zijdewind, Peter Feys, Felipe Balistieri Santinelli

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006949?dgcid=rss_sd_all

Interplay of action-based prediction and top-down attention: EEG evidence for joint modulation of late perceptual processing

📖 1 min 📝 16 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Thomas Holstein, Jean-Christophe Sarrazin, Bruno Berberian, Andrea Desantis

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006087?dgcid=rss_sd_all

Multivariate prediction of white matter BOLD activity and its relationship to marginal connectivity

📖 1 min 📝 15 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Muwei Li, Zhaohua Ding, John C Gore

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006129?dgcid=rss_sd_all

Amplitude-conditioned brain dynamics reveal widespread negative modulation and targeted positive modulation of functional networks

📖 1 min 📝 15 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Mohammad Reza Salehi, Ali BashirGonbadi, Hamid Soltanian-Zadeh

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006191?dgcid=rss_sd_all

Precise modeling of task-related sensorimotor activation based on simultaneous surface electromyography

📖 1 min 📝 22 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Monika Jasenská, Pavel Hok, Martin Kojan, Ondřej Burkot, Barbora Kolářová, Aleš Holobar, Petr Hluštík

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006154?dgcid=rss_sd_all

Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

Classic and novel features of the dentate gyrus vascular niche quantified through 3D vessel reconstruction

Carter, I., Smith, B. M., Kirby, E. D.

2026-10-24 1 min 331 words

BIORXIV NEUROSCIENCE

Summary: Adult neurogenesis, the continued generation of new neurons throughout life, supports a variety of emotional and cognitive functions in mammals. Lifelong neurogenesis occurs in the dentate gyrus (DG) of the hippocampus, where a neurogenic niche supports the survival and growth of neural stem cells (...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.20.760295v1?rss=1

Nanoscale remodeling of synaptic glutamate receptors and calcium channels in temporal lobe epilepsy

Kaufmann, W. A., Venturino, A., Kleindienst, D., Roessler, K., Czech, T., Janssen, J. M., Engelhardt, M., Stefanits, H., Watanabe, M., Siegert, S., Shigemoto, R.

2026-10-24 1 min 372 words

BIORXIV NEUROSCIENCE

Summary: Temporal lobe epilepsy (TLE) is the most common form of focal epilepsy and remains drug-resistant in approximately one-third of patients, for whom surgical resection of the epileptogenic zone is currently the most effective treatment option. One strategy to improve therapeutic outcomes involves iden...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.760440v1?rss=1

Limb-localized induction of regeneration reshapes brain and amputation-site transcriptional states in adult Xenopus laevis

Murugan, N. J., Russell, S., Kaplan, D., Levin, M., Miller, K.

2026-10-24 1 min 215 words

BIORXIV NEUROSCIENCE

Summary: Limb regeneration requires coordinated tissue reconstruction and nervous system reintegration, yet whether regenerative induction alters brain responses to limb loss remains unclear. We analyzed transcriptomics from the paired brains and amputation sites of adult Xenopus laevis after hindlimb amputa...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761414v1?rss=1

Muscarinic signaling sustains delay-period selectivity in prefrontal cortex by preserving thalamic drive

Kim, Y., Kim, J., Choi, I. S., Lee, J., Yang, Y.-S., Choi, J. H., Choi, J.-W., Rah, J.-C.

2026-10-24

1 min 164 words

BIORXIV NEUROSCIENCE

Summary: Short-term memory (STM) requires the temporary maintenance of task-relevant information in prefrontal cortex (PFC). Reciprocal interactions between the mediodorsal thalamus (MD) and PFC have been implicated in sustaining this activity, but how thalamofrontal communication is maintained despite the b...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.20.760355v1?rss=1

Engineered proteins to restore and enhance neurotransmitter release

👤 Chin, M., Yang, K., Lopez, R. C., Tan, C., Kaeser, P. S.

📅 2026-10-24 📖 1 min 📝 158 words

BIORXIV NEUROSCIENCE

Summary: Synapse dysfunction is a hallmark feature of many neurological diseases. Synaptic transmission is initiated in nerve terminals, where active zone protein complexes control vesicle fusion for neurotransmitter release. Here, we engineer proteins that mediate active zone functions and demonstrate their...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761409v1?rss=1

A genome-wide catalog of sites where peripheral DNA methylation reflects the brain

👤 Nishitani, S., Han, S., Ishii, T., Shimura, A., Phuong, N. J., Kato, A., Chambers, M., Wang, T., Isozaki, M., Higashino, Y., Yamauchi, T., Kawajiri, S., Arishima, H., Kodera, T., Imoto, H., Nomura, S., Yamanishi, K., Aoyama, B., Genc, I., Sakuma, R., Nguyen, H. D., Gorantla, N., Ghasemi, E., Yao, A., Kurata, S., Fujisawa, T. X., Braun, P. R., Nagao, T., Yamanashi, T., Tomoda, A., Kikuta, K., Ooka, T., Grant, G. A., Hooper, J. E., Howard, M. A., Kawasaki, H., Potash, J. B., Snyder, M. P., Shinozaki, G.

📅 2026-10-24 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

Summary: Epigenetic studies of psychiatric and neurological disorders rely largely on peripheral tissues, yet where peripheral DNA methylation (DNAm) reflects the brain remains unclear. We generated a paired brain and peripheral DNAm dataset from 148 donors in three cohorts and combined it with five publishe...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761276v1?rss=1

Sound-Driven Motion of Human Cochlear Structures Depends on Organ of Corti Integrity

👤 Secchia, P., McHugh, C. I., Cho, N. H., Zosuls, A., Tubelli, A., Puria, S., Nakajima, H. H.

📅 2026-10-24

📖 1 min 📝 153 words

BIORXIV NEUROSCIENCE

Summary: Optical coherence tomography (OCT) was used to image and measure the passive sound-driven displacements of cochlear structures in situ through the intact round window membrane of fresh human cadaveric specimens at 12 to 30 hours postmortem. We find that the anatomical integrity of the organ of Corti...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.19.759683v1?rss=1

Modelling prodromal Parkinson's disease- gastrointestinal pathology and early motor deficits in mice after transient rotenone exposure

👤 Selvaraji, S., Mosberger, J., Pang, K. K. L., Eberly, G. L., Kim, Y. J., Mondal, R., Leomi, R., Anikeeva, P.

📅 2026-10-24 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

Summary: Early gastrointestinal (GI) dysfunction is a hallmark feature of Parkinson's disease (PD), yet models capturing prodromal gut and brain pathology in PD remain limited. Motivated by the increased PD risk associated with pesticide exposure, we monitored motor behavior, brain pathology, and GI structur...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.761386v1?rss=1

Alzheimer's Disease-Associated Proteomic Changes in Olfactory Stem Cells Show Pathway-Level Transcriptomic Overlap

👤 Rantanen, L. M., Robinson, H., DutraVoigt, D., Diaz, X., Lampinen, R., Mykkanen, S., Kalapudas, J., Koivisto, A. M., Penttila, E., Oikari, L. E., Quek, H., Kanninen, K. M., White, A. R.

📅 2026-10-24 📖 1 min 📝 255 words BIORXIV NEUROSCIENCE

Summary: Alzheimer's disease (AD) is a progressive, neurodegenerative disorder and the leading cause of dementia worldwide. Although recent therapeutic advances have provided some disease-modifying potential, there is currently no curative treatment for AD, highlighting the need for improved disease models a...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.19.760270v1?rss=1

Brain pericytes are an oxidative stress-sensitive cell population damaged through ROS-1 dependent TRPA1 activation.

Groten, C. J., Kwan, J., Groening, J., Belleville, M., Weilinger, N., Bai, Y., Wendt, S., Kortje, Z., Olarinde, F., Chung, I., Choi, H. B., Bernier, L.-P., Rungta, R., MacVicar, B. A.

2026-10-24 1 min 200 words BIORXIV NEUROSCIENCE

Summary: Oxidative stress is a pathological process driven by an imbalance between the production and clearance of reactive oxygen species (ROS). It is a hallmark of numerous brain disorders, driving neuroinflammation, blood-brain barrier (BBB) disruption, and neurodegeneration. Here, we used two-photon micr...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.19.760199v1?rss=1

Altered brain-body activity during emotional mental imagery in major depressive disorder

2026-10-24 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41598-026-75968-1

Psychosocial stress is associated with lipidomic gut microbial and brain connectivity differences relevant to cardiometabolic vulnerability

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-75747-y

Physical exercise and esports performance: an updated systematic review and study-level meta-analysis

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-74442-2

Dynamic construction behavior recognition through domain-adaptive multimodal evidence fusion

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-74823-7

A synaptic mitochondrion–ribbon–vesicle complex organizes photoreceptor neurotransmission

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78869-z

Correction for Liu et al., A spinoreticular pathway mediates nocifensive responses to noxious mechanical stimuli

📅 2026-10-21 📖 1 min 📝 15 words PNAS NEUROSCIENCE

> **Summary:** Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2635736123?af=R

Compensation of the effects of temperature on a motor system in the crab, Cancer borealis

👤 Kathleen JacquerieJames M. DiMartinoAnanya DalalJi ZengEve Marderahttps://ror.org/05abbep66Volen Center and Biology Department, Brandeis University, Waltham, MA 02454

📅 2026-10-21 📖 1 min 📝 50 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceCold-blooded animals that live in climates with significant seasonal changes in ambient temperature must have myriad mechanisms to function over a wide range of environmental conditions. We explore...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625185123?af=R

Action features dominate cortical representation during natural vision

👤 Heejung JungHunar JainXiaochun HanMa FeilongJane HanDeepanshi ShokeenCara Van UdenIsabella HansonAndrew C. ConnollyTor D. WagerJames V. HaxbySamuel A. Nastaseahttps://ror.org/049s0rh22Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH 03755bIndependent Researcher, Sunnyvale, CA 94085chttps://ror.org/03taz7m60Department of Psychology, Center for Computational Language Sciences, University of Southern California, Los Angeles, CA 90089

📅 2026-10-23 📖 1 min 📝 51 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceHuman cortex is functionally organized to prioritize the most important features of our external world. In perceiving the natural world, however, these features are often overlapping and intermixed...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625018123?af=R

Translational control of microglial metabolism, inflammatory responses, and survival by the eIF4E-binding proteins

👤 Sara BermudezJung-Hyun ChoiNiaz MahmoodJacob W. VogelHashem AlmousaZiying HuangSung-Hoon KimVivian Yuchan ZhuLinqiao ZhouDiana Casas-AntillonMoein YaqubiDanielle CozachencoJo Anne StrattonOskar HanssonLuke HealyArgel Aguilar-VallesNahum Sonenbergahttps://ror.org/01pxwe438Department of Biochemistry and Goodman Cancer Institute, McGill University, Montreal, QC H3A 1A3, Canadabhttps://ror.org/02wnxgj78Department of Biochemistry, Chungbuk National University, Cheongju 28644, South Koreachttps://ror.org/012a77v79Department of Clinical Sciences, Lund University, Lund 221 84, SwedendSciLifeLab, Lund 221 00, Swedenehttps://ror.org/0227as991Department of Microbiology and Molecular Biology, Chungnam National University, Daejeon 34134, South Koreafhttps://ror.org/03rmrcq20Department of Pathology and Laboratory Medicine, Faculty of Medicine, University of British Columbia, Vancouver, BC V5Z 1M9, CanadagThe Hospital for Sick Children, Toronto, ON M5G 1E8, Canadahhttps://ror.org/01pxwe438Department of Neurology and Neurosurgery, Montreal Neurological Institute, McGill University, Montreal, QC H3A 1A1, Canadaihttps://ror.org/03490as77Institute of Biophysics Carlos Chagas Filho, Federal University of Rio de Janeiro, Rio de Janeiro 21941-170, Braziljhttps://ror.org/02qtvee93Department of Neuroscience, Carleton University, Ottawa, ON K1S 5B6, Canada

📅 2026-10-21 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceThe link between Alzheimer’s disease (AD) risk and microglial function is well-known, yet the mechanisms governing microglial responses remain unclear. Postmortem and experimental studies demonstra...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2610766123?af=R

Naturalistic spatial education engages distinct neural processes for mental rotation, cognitive load, and linguistic demands

👤 Eloise B. GacettaMarissa L. LawsLauren S. AuletTeoman S. OzaydinJonathan S. TsayConrad ZapantaJessica F. Cantlonahttps://ror.org/05x2bcf33Department of Psychology, Carnegie Mellon University, Pittsburgh, PA 15213bhttps://ror.org/01an3r305Center for the Neural Basis of Cognition, Carnegie Mellon University and University of Pittsburgh, Pittsburgh, PA 15213chttps://ror.org/0072zz521Department of Psychological and Brain Sciences, University of Massachusetts Amherst, Amherst, MA 01003dhttps://ror.org/05x2bcf33Neuroscience Institute, Carnegie Mellon University, Pittsburgh, PA 15213ehttps://ror.org/05x2bcf33Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA 15213

📅 2026-10-21 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceSpatial visualization skill is a foundational component of human intelligence and a powerful lever for success in science, technology, engineering, and mathematics (STEM). Yet little is known about...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2601206123?af=R

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

Chun-Yan WangWei-Zhu LiuShuang-Mei LiaoJing NiuHan-Qing PanWen-Jie YouKe ZhaoBao MaMing-Fen LuJia-Xin ZouPing HuWen-Hua ZhangBing-Xing Panahttps://ror.org/042v6xz23Jiangxi Institute of Respiratory Diseases and Department of Pathology, The First Affiliated Hospital, School of Basic Medical Sciences and Institute of Biomedical Innovation, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinabhttps://ror.org/042v6xz23Jiangxi Provincial Key Laboratory of Brain Science and Brain Health, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinachttps://ror.org/042v6xz23Postdoctoral Research Station, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinadhttps://ror.org/042v6xz23Institute of Translational Medicine, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinaehttps://ror.org/042v6xz23Department of Neurology, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, China

📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

👤 Tingwei Quan 📅 2026-10-07 📖 1 min 📝 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

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Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

👤 Bocheng Wang 📅 2026-10-07 📖 1 min 📝 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1869687

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

👤 Ansel Y. Rodríguez-González
📅 2026-10-07 📖 1 min 📝 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

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Mechanonociceptive tactile feedback enhances prosthetic grasp control during repeated human-in-the-loop interaction

👤 Elif Hocaoglu
📅 2026-10-15 📖 1 min 📝 215 words

FRONTIERS NEUROROBOTICS

Summary: Upper-limb prosthetic systems continue to face a fundamental challenge in providing sensory feedback that remains functionally effective during repeated interaction. This study investigates innocuous painful feedback (IPF), a controlled mechanonociceptive tactile modality intended to provide salient...

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Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han
2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

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Single-channel EEG denoising and artifact removal: a systematic scoping review of methods, benchmarking, and reproducibility

Usman Qamar Shaikh, Anubha Manju Kalra, Andrew Lowe and Imran Khan Niazi
2026-10-14

1 min 239 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Single-channel and wearable electroencephalography (EEG) reduce acquisition burden but limit the spatial information available to conventional multichannel artifact-removal methods. We conducted a systematic scoping review of single-channel-compatible EEG artifact removal, linking method ...

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Phase-locked closed-loop ultrasound stimulation of hippocampal slow gamma modulates CA3-CA1 coherence and improves spatial memory in APP/PS1 mice

👤 Zhenyu Xie, Hui Ji, Mürsel Karadas, Yiyao Zhang and Yi Yuan

📅 2026-10-14 📖 1 min 📝 235 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Impaired hippocampal slow gamma oscillations and CA3-CA1 dysfunction may contribute to spatial memory deficits in early Alzheimer's disease (AD). This study examined whether phase-locked closed-loop transcranial ultrasound stimulation (TUS) could modulate CA3-CA1 activity and improve memo...

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Ordered onset of regional desynchronization and a transition in movement-class discriminability during cue-paced motor imagery

👤 Seho Lee

📅 2026-10-21 📖 1 min 📝 298 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor imagery recruits a distributed network whose components and large-scale organization are well characterized. The temporal order in which those components are engaged is not: the latency at which each region begins to desynchronize within a trial has not been compared statistically i...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb161

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ultrabright Quasi-Compact Quantum Dot Assembly for Sensitive and Color-Fixed Immunoassays

👤 Liang Huang
📅 2026-10-23 📖 1 min 📝 63 words

LOW VISION

Summary: Home-based immunoassay platforms constitute critical tools for the rapid screening of infectious respiratory viruses. Notably, viral loads can remain low at certain stages yet vary over several orders of magnitude during infection, and such variations are closely associated with the disease progress...

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Supercritical Carbon Dioxide-Peracetic Acid Treatment Inactivates Low-Level Bacterial Inocula in Collagen Analogue-Phosphorylcholine Hydrogels while Preserving Functional Properties

👤 May Griffith
📅 2026-10-23 📖 1 min 📝 57 words

LOW VISION

Summary: Terminal sterilization of hydrogels is challenging because their high-water-content, chemical structures, and cell-friendly properties are often altered by conventional sterilization methods. Hydrogel implants comprising collagen-like peptide (CLP), polyethylene glycol (PEG), and poly (2-methacryloy...

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A Bidirectional Photosensitive Multimodal PdSe2/MoS2 van der Waals Heterostructure Memristor Array for In-Sensor Neuromorphic Vision

👤 Jun He 📅 2026-10-23 📖 1 min 📝 60 words LOW VISION

Summary: Optoelectronic van der Waals (vdW) memristors integrating photosensing, memory, and computing capabilities are promising for artificial vision but are limited by single photoresponse modes, low yields, and high device variability. Here, we demonstrate a 1-kb PdSe2/MoS2 van der Waals heterostructure ...

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Al-Stoichiometry-Engineered Defect Profiles in InAlZnO Nanofibers for Balanced Charge Transport and Optical Plasticity

👤 Fengyun Wang
📅 2026-10-23 📖 1 min 📝 64 words

LOW VISION

Summary: Metal oxide optoelectronic synapses are promising for neuromorphic vision, but combining a low dark current with sustained photoinduced plasticity remains challenging. Here, Al stoichiometry is engineered in electrospun In1AlxZn0.08O (x = 0.007, 0.011, and 0.015) nanofiber networks to tune sub-bandg...

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Effect of 0.01%, 0.02% and 0.04% Atropine eyedrops vs Placebo on myopia incidence in pre-myopic children: A randomized clinical trial

Xiangui He 2026-10-23 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Low-concentration atropine demonstrated a dose-response relationship for reducing cumulative myopia incidence and slowing refractive progression in pre-myopic children, but no such relationship on axial length. This preliminary evidence suggests potential for early myopia prevention, bu...

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Diagnosis of a chiasmal lesion compressing optic chiasm in patients with severe retinal degeneration: a two-case report

Kyung-Ah Park 2026-10-23 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Optic chiasmal compression should be considered in patients with severe retinal degeneration presenting with progressive visual decline that cannot be fully explained by retinal pathology. Visual field testing and timely neuroimaging, particularly MRI, are essential for accurate diagnos...

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Reading Arabic Under Simulated Low Vision: Orthographic Complexity Shapes the Effect of Contrast Polarity

👤 Mosaad Alhassan
📅 2026-10-23 📖 1 min 📝 45 words

LOW VISION

Summary: CONCLUSION: Arabic relies on small, high-spatial-frequency features such as dots for letter discrimination, which appear to modulate how contrast polarity interacts with optical degradation at threshold. These findings provide preliminary evidence that may guide future studies of reading performance...

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Screening-based prevalence and axis distribution of astigmatism among children and adolescents in Lintong District, Xi'an: a large-scale school-based cross-sectional study

Zhaojiang Du
2026-10-23 1 min 51 words LOW VISION

Summary: CONCLUSIONS: Astigmatism was common in this large district school screening dataset and differed by school stage, axis type, and refractive-status category. The findings may help local screening programs plan re-examination and referral pathways, but they should not be interpreted as cycloplegic cli...

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Genome mining and comparative genome analysis of Pseudomonas fluorescens revealed diverse biosynthetic and metabolic potential

👤 Farida Begum 📅 2026-10-24 📖 1 min 📝 70 words LOW VISION

Summary: Bacterial genome sequencing has been used to identify a myriad of bioactive compounds that have yet to be characterised. Pseudomonas fluorescens is a widely distributed Gram-negative bacterium known for its plant growth-promoting traits and production of diverse secondary metabolites. P. fluorescens...

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Clinical profiles, referral patterns and low vision aid prescribing across four rehabilitation centres in Ghana: protocol for a multicentre retrospective record review

Werner Eisenbarth 2026-10-24 1 min 79 words LOW VISION

Summary: BACKGROUND: Low vision, defined as best-corrected visual acuity of less than 6/18 in the better eye or a visual field constriction of 20° or less, is a condition in which residual vision may be optimized through optical and non-optical aids. Despite the importance of low vision services in maximizin...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

Xin Zhou 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

Peng Chen 2026-10-23 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

Xiaojun Zhou 2026-10-23 1 min 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

👤 Hussein Youssef 📅 2026-10-23 📖 1 min 📝 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

👤 Leticia González 📅 2026-10-23 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Nonpharmacological Interventions in Degenerative Cerebellar Ataxia: A Systematic Review

👤 Pramod Kumar Pal 📅 2026-10-23 📖 1 min 📝 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: NPI may confer short-term symptomatic and functional benefits in degenerative cerebellar ataxia, particularly in axial motor domains, supported by the largest evidence for structured rehabilitation followed by neuromodulation, although heterogeneous. Future multicenter, biomarker-inform...

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Effects of Orbitofrontal Cortex tDCS on Cognitive Control in Individuals With Obesity: A Randomized Sham-Controlled Trial

👤 Stephanie L Borgland 📅 2026-10-23 📖 1 min 📝 43 words TDCS TACS TRNS

Summary: CONCLUSIONS: Unlike what has been reported in lean individuals, a single session of OFC-targeted tDCS is insufficient to modify cognitive control in individuals with obesity. However, the cognitive effects of a single session of OFC stimulation depend on individual neurocognitive characteristics in ...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

Hua Hu
2026-10-21 1 min 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

👤 Balaji Jayaprakash 📅 2026-10-22 📖 1 min 📝 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

👤 Jing Zhang 📅 2026-10-23 📖 1 min 📝 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

👤 Rihui Li 📅 2026-10-23 📖 1 min 📝 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

👤 Esra Suzen 📅 2026-10-23 📖 1 min 📝 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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The test-retest reliability of cortical activation and behavioral metrics in adolescent basketball players during dual-task postural control: an fNIRS-based assessment

👤 Lulu Song 📅 2026-10-24 📖 1 min 📝 59 words FNIRS

Summary: CONCLUSION: These findings provide preliminary test-retest reliability estimates for selected multimodal neurobehavioral measures in male adolescent basketball players although reliability alone does not establish clinical utility and estimates with wide confidence intervals require confirmation in ...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

Qingguo Wei 2026-10-23 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

👤 Jiayuan He 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Training Free Signal Enhancement by Random Sampling for Steady State Visual Evoked Potential BCI

👤 Yoshikazu Washizawa 📅 2026-10-23 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: The steady-state visual evoked potential (SSVEP) is widely used for
electroencephalography (EEG)
based brain computer interfaces (BCIs).
In order to determine the target commands for SSVEP-BCIs,
deep neural networks have been widely used.
However, learning-based approaches requir...

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Decoding attempted handwritten characters based on temporal endovascular neural features

Peter E Yoo 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: We evaluated temporal neural features for motor decoding using a minimally-invasive endovascular brain-computer interfaces (BCI). We hypothesised that relatively long compound movements generate distinct patterns of neural activity that are encoded by temporal features.

Approach. A single p...

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Transformer networks are robust learners of source localization for EEG measurements

Moritz Dannhauer 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

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Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

Chen Wang 2026-10-23 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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Generalizable neural decoding for clinically scalable brain-computer interfaces

👤 Jie Chen 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces promise restored communication and movement, yet accuracy achieved under development rarely survives new sessions, users, tasks, or hardware. Generalization, not peak performance, determines clinical scalability. Here we examine generalizable neural decoding from developmen...

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Polygenicity shapes distinct brain patterns and clinical profiles of bipolar disorder

👤 Shaohua Hu 📅 2026-10-23 📖 1 min 📝 41 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Our findings reveal associations between polygenicity and distinct BD brain patterns and clinical profiles. We identified ancestry-specific causal variants and bidirectional BD-IDP relationships. PRS associations with depressive symptoms, anxiety, suicide risk, and metabolic phenotypes ...

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Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

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🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

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https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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https://fmhy.net/posts/sept-2026

[CVE-2026-111497] If you run a python server, you should update it right now

👤 /u/me_myself_ai 📅 2026-10-24 📖 1 min 📝 296 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> The "HTTP2Shell" vulnerability affects servers that know how to speak Google's 'brotli' compression protocol, which would likely make its way into your server through an innoccuous-sounding speedup dep like <code>urllib3[brotli]</code> or <code>aiohttp[spe...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xe1zp2/cve2026111497_if_you_run_a_python_server_you/

Hacker reverses Samsung TV, finds on-board AI listening OpenAI account suspended

👤 testing22321 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.reddit.com/r/antiai/comments/1xdv3e5/hacker_reverse_engineers_his_samsung_tv_and_finds/ Comments URL: <a href="https://news.ycombinator.com/item?id=502...

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https://www.reddit.com/r/antiai/comments/1xdv3e5/hacker_reverse_engineers_his_samsung_tv_and_finds/

Node2Shell evolves to HTTP2Shell, blast radius affects many HTTP libs and apps

👤 theanonymousone 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.aikido.dev/blog/python-packages-vulnerable-node2shell-brotli-cve-2026-111497 Comments URL: <a href="https://news.ycombinator.com/item?id=50219177">https://news.yc...

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I Don't Understand Quantum Physics (2018)

👤 rramadass 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.southampton.ac.uk/~doug/quantum_physics/index.html Comments URL: https://news.ycombinator.com/item?id=50219666 Points: 87...

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Node.js Vulnerability Blog

👤 Brysonbw 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nodejs.org/en/blog/vulnerability Comments URL: https://news.ycombinator.com/item?id=50220145 Points: 6 # Comments: 1

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https://nodejs.org/en/blog/vulnerability

AI-Assisted Paper on Z-Library Cites a TorrentFreak Article That Doesn't Exist

👤 TangerineDream 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://torrentfreak.com/ai-assisted-paper-on-z-library-cites-a-torrentfreak-article-that-doesnt-exist/ Comments URL: <a href="https://news.ycombinator.com/ite...

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New software package: ERPLAB Studio

Steve Luck 2024-06-12 2 min 449 words ERP BOOT CAMP

Summary: <p class="">We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it's a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

2026-01-16 1 min 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11230544

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11319164

A critical assessment of LGN segmentation quality based on volume, contour, and location

👤 Schroeder, J.-L., Gulban, F. O., Mueller-Axt, C., Goebel, R., Kurzawski, J. W.

📅 2026-10-23 📖 1 min 📝 207 words BIORXIV NEUROSCIENCE

Summary: Reliable segmentation of the lateral geniculate nucleus (LGN), a small thalamic relay nucleus essential for visual processing, is necessary for characterizing its anatomy in healthy populations and structural alterations in neurological disorders. However, accurate delineation remains challenging on...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.19.759668v1?rss=1

Endogenous retrovirus activation disrupts postsynaptic organization in human glutamatergic neurons

Davis-Hansson, C., Garza, R., Piochon, C., Johansson, J. G., Busskamp, V., Jakobsson, J., Johansson, E. M.

2026-10-23 1 min 141 words BIORXIV NEUROSCIENCE

Summary: Endogenous retroviruses (ERVs) comprise a substantial fraction of the human genome, yet whether their expression can directly alter human neuronal function remains unknown. Because glutamatergic signaling is fundamental to brain function and frequently disrupted in brain disorders, we asked whether ...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760639v1?rss=1

Human and Mouse Models of the Bipolar Disorder-Associated ANK3 p.W1989R Variant Display Altered Somatostatin Interneuron Connectivity and Function

👤 Dean, K. P., Verduci, S. C., Kramm, C. C., Stinson, C. W., Min, L., Smith, J. S., DeLong, C. J., Yao, M., Lee, K. H., Reeves, J. M., Birdsong, W. T., Welch, J. D., Jenkins, P. M.

📅 2026-10-23 📖 1 min 📝 284 words BIORXIV NEUROSCIENCE

Summary: Bipolar disorder (BD) is a debilitating neuropsychiatric disorder characterized by recurrent mood perturbations, cognitive and functional impairment, and increased risk of suicide. Genome-wide association studies (GWAS) have repeatedly identified the gene ANK3, which encodes the neuronal scaffolding...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.23.760718v1?rss=1

Pediatric brain charts reveal ADHD biotypes with distinct developmental, molecular, and therapeutic profiles

👤 Bu, X., Sun, L., Fu, Z., Zhao, Y., Zheng, X., Zhang, K., Liang, X., Cui, Z., Gao, J.-H., Gong, G., Huang, R., Li, S., Luo, L., Men, W., Peng, Y., Sun, L., Sun, X., Wang, X., Xia, M., Zhao, T., Zhu, L., Zuo, X., Wang, Y., Dong, Q., Cao, Q., Yang, L., He, Y.

📅 2026-10-23 📖 1 min 📝 183 words BIORXIV NEUROSCIENCE

Summary: Attention-deficit/hyperactivity disorder (ADHD) exhibits profound clinical heterogeneity; however, its underlying neurobiological substrates remain unclear. Here, leveraging large - scale structural magnetic resonance imaging data from 12,663 typically developing children and 1,699 children with ADH...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.22.760978v1?rss=1

Unstable word form representations and greater reliance on right hemisphere during word learning in adults with a history of developmental language disorder

Nora, A., Yan, C., Husa, V., Wilkki, V., van Vliet, M., Service, E., Salmelin, R.

2026-10-23 1 min

205 words

BIORXIV NEUROSCIENCE

Summary: Impairment in acquiring new vocabulary is a persistent deficit in developmental language disorder (DLD), but its cortical substrates remain unclear. We investigated learning of new words in young adults with DLD and typically developing (TD) control participants using magnetoencephalography (MEG). P...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.21.760607v1?rss=1

Predictive dendrites as a foundation for biological learning

Song, Y., Innocenti, F., Mirkhani, N., Bozkurt, B., Ivan, M., Aziz, A., Lukasiewicz, T., King, A. J., Clopath, C., Akerman, C., Harper, N. S., Bogacz, R.

2026-10-23 1 min

152 words

BIORXIV NEUROSCIENCE

Summary: A fundamental challenge for neuroscience is to identify laws of synaptic plasticity that are consistent with experimental observations at the level of individual neurons, whilst also explaining the ability of neural networks to learn complex tasks. Here we investigate a candidate law in which neuron...

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https://www.biorxiv.org/content/10.64898/2026.10.21.760638v1?rss=1

Deficiency of non-muscle myosin IIA promotes neuroinflammation by impairing autophagic degradation of VCAM-1 in brain endothelium

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-75768-7

Risk model for predicting mortality risk in patients with cerebral infarction: a cohort study using the MIMIC-IV database

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-76494-w

Enterovirus A71 exploits peripheral circulating extracellular vesicles to facilitate blood-brain barrier permeability

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-79119-y

Inhibition of spinal astrocytic GPR177 attenuates neuropathic pain by modulating C3/C3aR signaling in male rats

📅 2026-10-24 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11221-9

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Action features dominate cortical representation during natural vision

👤 Samuel A Nastase 📅 2026-10-23 📖 1 min 📝 70 words LOW VISION

Summary: Cortical resources are allocated into systems specialized for processing ecologically relevant features of the world. These features are typically studied in isolation using controlled experimental stimuli, making it difficult to assess the relative importance of different kinds of features that ten...

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Beyond the browning index: RGB radiomics enables non-destructive prediction of fresh-cut "Red Fuji" apple browning and enzyme activities

👤 Tian Mou 📅 2026-10-23 📖 1 min 📝 62 words LOW VISION

Summary: Fresh-cut apples undergo rapid enzymatic browning, which compromises visual quality and causes substantial postharvest losses. Most RGB vision approaches rely on handcrafted color/texture descriptors that are sensitive to acquisition conditions and provide limited linkage to the underlying enzyme dy...

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Discovery, Design, and Evaluation of Nanomolar to Picomolar Inhibitors of Human Pancreatic α-Amylase for Control of Postprandial Blood Glucose Levels

Stephen G Withers 2026-10-23 1 min 80 words LOW VISION

Summary: Type 2 diabetes is a disease that is rapidly expanding across the world with global cases quadrupling since 1990 and now comprising over 800 million people, the majority of whom are in low- and middle-income countries (Zhou et al.Lancet2024, 404, 2077-2093.). The poor control of blood glucose levels...

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High Performance Switchable Dual-Mode Photoresponse Enabled by Defect States in ReS2/PdSe2 Heterojunctions

Zhesheng Chen 2026-10-23 1 min 65 words LOW VISION

Summary: Two-dimensional (2D) semiconductors exhibit significant potential for next-generation optoelectronic and neuromorphic applications. In particular, intrinsic defect states in 2D materials not only modulate the doping level but also play key roles in the photoresponse time of optoelectronic devices. D...

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Ultrabright Quasi-Compact Quantum Dot Assembly for Sensitive and Color-Fixed Immunoassays

👤 Liang Huang
📅 2026-10-23 📖 1 min 📝 63 words

LOW VISION

Summary: Home-based immunoassay platforms constitute critical tools for the rapid screening of infectious respiratory viruses. Notably, viral loads can remain low at certain stages yet vary over several orders of magnitude during infection, and such variations are closely associated with the disease progress...

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Supercritical Carbon Dioxide-Peracetic Acid Treatment Inactivates Low-Level Bacterial Inocula in Collagen Analogue-Phosphorylcholine Hydrogels while Preserving Functional Properties

👤 May Griffith
📅 2026-10-23 📖 1 min 📝 57 words

LOW VISION

Summary: Terminal sterilization of hydrogels is challenging because their high-water-content, chemical structures, and cell-friendly properties are often altered by conventional sterilization methods. Hydrogel implants comprising collagen-like peptide (CLP), polyethylene glycol (PEG), and poly (2-methacryloy...

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A Bidirectional Photosensitive Multimodal PdSe2/MoS2 van der Waals Heterostructure Memristor Array for In-Sensor Neuromorphic Vision

👤 Jun He 📅 2026-10-23 📖 1 min 📝 60 words LOW VISION

Summary: Optoelectronic van der Waals (vdW) memristors integrating photosensing, memory, and computing capabilities are promising for artificial vision but are limited by single photoresponse modes, low yields, and high device variability. Here, we demonstrate a 1-kb PdSe2/MoS2 van der Waals heterostructure ...

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Al-Stoichiometry-Engineered Defect Profiles in InAlZnO Nanofibers for Balanced Charge Transport and Optical Plasticity

👤 Fengyun Wang
📅 2026-10-23 📖 1 min 📝 64 words

LOW VISION

Summary: Metal oxide optoelectronic synapses are promising for neuromorphic vision, but combining a low dark current with sustained photoinduced plasticity remains challenging. Here, Al stoichiometry is engineered in electrospun In1AlxZn0.08O (x = 0.007, 0.011, and 0.015) nanofiber networks to tune sub-bandg...

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Effect of 0.01%, 0.02% and 0.04% Atropine eyedrops vs Placebo on myopia incidence in pre-myopic children: A randomized clinical trial

Xiangui He 2026-10-23 1 min 47 words LOW VISION

Summary: CONCLUSIONS: Low-concentration atropine demonstrated a dose-response relationship for reducing cumulative myopia incidence and slowing refractive progression in pre-myopic children, but no such relationship on axial length. This preliminary evidence suggests potential for early myopia prevention, bu...

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Diagnosis of a chiasmal lesion compressing optic chiasm in patients with severe retinal degeneration: a two-case report

Kyung-Ah Park 2026-10-23 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Optic chiasmal compression should be considered in patients with severe retinal degeneration presenting with progressive visual decline that cannot be fully explained by retinal pathology. Visual field testing and timely neuroimaging, particularly MRI, are essential for accurate diagnos...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

👤 Béatrice Berçot 📅 2026-09-04 📖 1 min 📝 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

👤 Xiaobing Yan 📅 2026-09-09 📖 1 min 📝 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

👤 Aliaa Elabd 📅 2026-10-20 📖 1 min 📝 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

Peng Chen 2026-10-23 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

Xiaojun Zhou 2026-10-23 1 min 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Hemispheric lesion asymmetry - An overlooked variable in tDCS targeting for multiple sclerosis

👤 Hussein Youssef 📅 2026-10-23 📖 1 min 📝 2 words TDCS TACS TRNS

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Gradient-Based Curvature-Only Time-Derivative Coupling for Surface Hopping and Coherent Switching with Decay of Mixing: Application to SO2 and Molecular Tully Models

👤 Leticia González 📅 2026-10-23 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: Time-derivative couplings (TDCs) are central ingredients in mixed quantum-classical simulations of nonadiabatic dynamics, yet their evaluation remains a key practical challenge when analytic nonadiabatic coupling vectors or wave function overlaps are unavailable. Curvature-driven approximations prov...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

Xuqun You
2026-10-22 1 min 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

👤 Rihui Li 📅 2026-10-23 📖 1 min 📝 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Local and network-level prefrontal reactivity to intermittent theta burst stimulation: a randomized, sham-controlled iTBS/fNIRS study

👤 Georg S Kranz 📅 2026-10-23 📖 1 min 📝 2 words FNIRS

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Correction-stage information leakage and target-input asymmetry can distort comparisons of fNIRS short-separation channel correction strategies

Esra Suzen 2026-10-23 1 min 16 words FNIRS

Summary: OBJECTIVE: We examined how evaluation design affects comparisons of functional near-infrared spectroscopy (fNIRS) short-separation correction strategies.

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Conductive hydrogel bioelectronics for chronic neuromodulation: From biomimetic design and bioactive interfaces to integrated neural platforms

Xulin Hu 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: In biomedical engineering, hydrogel bioelectronics brings together the interface between flexible electronic devices and biological systems, opening up a new way for chronic neural regulation. Here we focus on the recent progress in this field, from material design to system integration. The design ...

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Adaptive local temporal weighting for short-window EEG-based SSVEP recognition using FB-eTRCA

Zehua Wang 2026-10-23 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: To improve short-window SSVEP recognition, we integrated adaptive temporal local weighting (ATLW) with filter-bank ensemble task-related component analysis (FB-eTRCA). ATLW estimates local discriminability from training trials, converts window scores into sample weights, and applies the training-der...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

👤 Qingguo Wei 📅 2026-10-23 📖 1 min 📝 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

👤 Jiayuan He 📅 2026-10-23 📖 1 min 📝 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Training Free Signal Enhancement by Random Sampling for Steady State Visual Evoked Potential BCI

Yoshikazu Washizawa 2026-10-23 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: The steady-state visual evoked potential (SSVEP) is widely used for
electroencephalography (EEG)
based brain computer interfaces (BCIs).
In order to determine the target commands for SSVEP-BCIs,
deep neural networks have been widely used.
However, learning-based approaches requir...

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Decoding attempted handwritten characters based on temporal endovascular neural features

Peter E Yoo 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: We evaluated temporal neural features for motor decoding using a minimally-invasive endovascular brain-computer interfaces (BCI). We hypothesised that relatively long compound movements generate distinct patterns of neural activity that are encoded by temporal features.

Approach. A single p...

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Transformer networks are robust learners of source localization for EEG measurements

Moritz Dannhauer 2026-10-23 1 min 65 words BRAIN COMPUTER INTERFACE

Summary: EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade under realist...

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DNA metabarcoding reveals the trophic ecology of Critically Endangered juvenile green sawfish (Pristis zijsron) and sympatric predators

Justin D Brookes 2026-10-23 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: North-western Australia is home to many of Australia's rare and threatened elasmobranchs. Basic ecological data is still lacking for many of these species, including empirical dietary information. DNA metabarcoding was performed on cloacal swabs to obtain the first empirical insights into the trophi...

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Biomimetic Nanofluidic Synapse with Programmable DNA Ion Gating for Neuromorphic Sensing and Computing

👤 Chen Wang 📅 2026-10-23 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

Summary: Biological synapses process information through coupled neurotransmitter and ion fluxes, while conventional electron-based solid-state devices struggle to replicate this chemoelectrical signaling mechanism. Herein, we develop a biomimetic nanofluidic synapse designed for precise and programmable che...

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A research on the typical images of animals in the dreams of Naxi.

📅 2026-09-10 📖 1 min 📝 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

📅 2026-05-18 📖 1 min 📝 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000346

Offline webcam login monitor for Windows (photographs failed logins, local face recognition, alerts)

👤 /u/Ed0x86 📅 2026-10-24 📖 1 min 📝 134 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Here is a Windows tool idea: it takes a burst of webcam photos when someone logs in, unlocks, or fails a password attempt on my PC, and alerts me via email or telegram if it sees a face it doesn't know (it's AI local model face recognition). Feel free to fork...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xe0ury/offline_webcam_login_monitor_for_windows/

One Man's Pan Am 747 Replica [video]

👤 Teever 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

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https://www.youtube.com/watch?v=vAFwHGGfPz4

Twelve years of distributed SQLite

👤 otoolep 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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Alzheimer's breakthrough: 2 hrs of sleep restored without clearing brain plaques

👤 amichail 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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Apple Security Research Device

👤 neom 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

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https://security.apple.com/research-device/

How Block Blast harvests its players for ad money

👤 motionlessveloc 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ayoships.com/breakdowns/block-blast Comments URL: https://news.ycombinator.com/item?id=50217946 Points: 14 # Comments: 2

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How close can pure functional programming get to C?

👤 stmonty 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://rtfeldman.com/fp-vs-c Comments URL: https://news.ycombinator.com/item?id=50218195 Points: 51 # Comments: 6

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OBS Studio 33

👤 pentagrama 📅 2026-10-24 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/obsproject/obs-studio/releases/tag/33.0.0 Comments URL: https://news.ycombinator.com/item?id=50218430 Points: 12 ...

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https://github.com/obsproject/obs-studio/releases/tag/33.0.0

A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004567?dgcid=rss_sd_all

TEMPORARY REMOVAL: Corrigendum to “LPI alleviates Alzheimer’s disease pathology via the GPR55 receptor” [Neuroscience 611 (2026) 99–110]

📖 1 min 📝 26 words NEUROSCIENCE JOURNAL

Summary: Publication date: 7 December 2026 Source: Neuroscience, Volume 618 Author(s): Wenjie Xu, Jiaqi Cao, Yibo Liu, Zimo Wei, Xu Zha, Shumin Xie, Xicheng Liu, Wei Wang, Chen Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006287?dgcid=rss_sd_all

Differential value processing in the rostral and causal subthalamic nucleus during flexible and habitual decision-making in primates: Implications for cognitive dysfunction in Parkinson's disease

Lee, K., Lee, M., Jiang, H., Kim, H. F.
2026-10-23 1 min 222 words

BIORXIV NEUROSCIENCE

Summary: The subthalamic nucleus (STN) is a major target region of deep brain stimulation (DBS) for Parkinson's disease (PD), effectively alleviating motor symptoms. However, STN-DBS can also induce cognitive impairments, suggesting that the STN contributes to cognitive functions beyond simple motor control....

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https://www.biorxiv.org/content/10.64898/2026.10.19.760032v1?rss=1

Behavioral genomics identifies Creld1 as a cross-modal regulator of somatosensation

Bailong Xiao
2026-10-22 1 min 42 words

NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02484-w Jiang et al. developed a robust postnatal knockout platform for somatosensory genomics and behavioral screening, identifying Creld1 as a cross...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02484-w

Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-10-22 1 min 40 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 22 October 2026; doi:10.1038/s41593-026-02486-8 The dimensionality of intrinsic brain connectivity systematically shapes how task representations compress and expand across the cortex, and a...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02486-8

Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang 2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1938329

Retraction: Intervention effect of rehabilitation robotic bed under machine learning combined with intensive motor training on stroke patients with hemiplegia

📅 2026-10-14 📖 1 min 📝 0 words FRONTIERS NEUROROBOTICS

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.2023207

Overcoming subject variability in P300 decoding based on quantification by deep learning

👤 Fred R. Beyette 📅 2026-10-22 📖 1 min 📝 267 words FRONTIERS NEUROSCIENCE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1940267

Striatal and orbitofrontal grey matter volumes are associated with cognitive event related potentials during associative learning task performance in male and female children and adolescents

👤 Almira Kustubayeva 📅 2026-10-22 📖 1 min 📝 303 words FRONTIERS NEUROSCIENCE

Summary: IntroductionThe individual differences in gray matter volumes and event-related potentials may underlie variability in associative learning. However, little is known about the sex differences in these structure–function relationships across development.MethodsThe current study investigates associati...

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Multiparametric MRI radiomics and machine learning for outcome prediction in acute ischemic stroke: a retrospective study of 353 patients

👤 Tingting Li 📅 2026-10-22 📖 1 min 📝 285 words FRONTIERS NEUROSCIENCE

Summary: BackgroundAcute ischemic stroke (AIS) remains a leading cause of death and disability worldwide. Early prediction for clinical outcomes is essential for guiding therapy. Multiparametric Magnetic Resonance Imaging (MRI) offers detailed characterization of ischemic tissue and may predict prognosis in ...

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Continuous performance tests in the assessment of children with attention-deficit/hyperactivity disorder: a narrative review

Xiaoke Zhao 2026-10-22 1 min 231 words FRONTIERS NEUROSCIENCE

Summary: Attention-deficit/hyperactivity disorder (ADHD) is diagnosed primarily through clinical interviews, behavioral observations, and rating scales, while continuous performance tests (CPTs) provide complementary, performance-based measures of sustained attention, response inhibition, and response consis...

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Microglia in the neurogenic niche: emerging roles of immunometabolism

Yan Liu 2026-10-22 1 min 136 words FRONTIERS NEUROSCIENCE

Summary: Adult neurogenesis is dynamically regulated by the neurogenic niche, yet the mechanisms governing its remodeling are not fully elucidated. Current research has increasingly extended beyond the intrinsic properties of neural stem cells to consider the broader microenvironment in which neurogenesis oc...

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The mammalian regulatory architecture: biobehavioral portals and the organization of physiological state

Stephen W. Porges 2026-10-22 1 min 214 words FRONTIERS NEUROSCIENCE

Summary: IntroductionNeuromodulatory interventions can produce overlapping adaptive outcomes despite engaging substantially different biological pathways. Pathway-centered approaches identify how interventions initially engage the nervous system and characterize pathway-specific mechanisms. Convergence acros...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

👤 Yolima Baena 📅 2026-10-20 📖 1 min 📝 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

Luca Battaglini 2026-10-21 1 min 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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New challenges in horticultural IPM: artificial intelligence can enhance the use of biocontrol agents

Andrea Lovazzano 2026-10-22 1 min 59 words LOW VISION

Summary: Modern horticulture faces the dual imperatives of increasing global food production while mitigating the severe environmental and health impacts of conventional chemical pesticides. Biological control agents (BCAs), which utilize beneficial microorganisms, macroorganisms, semiochemical and botanical...

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Bridging the performance-efficiency gap: a resource-efficient hybrid DL-ML framework for jackfruit disease detection

👤 Hezerul Bin Abdul Karim
📅 2026-10-22 📖 1 min 📝 62 words LOW VISION

Summary: Complex environment variability and limited computational resources in agricultural settings create challenges in automated detection in tropical regions, where jackfruit diseases pose major threats to harvest productivity and food security. Conventional machine learning models often fail to general...

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Atomic Defect-Mediated Charge Trapping Enables Multimodal Plasticity in van der Waals Heterostructures for In-Sensor Neuromorphic Vision

👤 Zhiqun Lin
📅 2026-10-22 📖 1 min 📝 60 words LOW VISION

Summary: Neuromorphic hardware that unifies sensing, memory, and learning at the device level remains challenging due to limited optoelectronic comodulation and rigid architectures. Here, we report a flexible, van der Waals (vdW)-integrated MoS2/hexagonal boron nitride (h-BN)/graphene memtransistor in which ...

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Low-Dimensional Chiral Perovskites for Polarization-Sensitive Photodetection

Zhiyong Fan 2026-10-22 1 min 57 words LOW VISION

Summary: Emerging low-dimensional chiral perovskites have recently demonstrated significant application potential in circularly polarized light detection, spintronics, and quantum information technology due to their unique chiroptical activity and spin-selective properties. This review systematically summari...

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Visual context influences lateral balance while walking on winding paths

Jonathan B Dingwell 2026-10-22 1 min 28 words LOW VISION

Summary: BACKGROUND: Navigating daily environments necessitates adapting to ever-changing conditions while maintaining stability. This requires integrating sensory feedback, primarily from vision, to maintain balance and maneuverability as we walk.

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Bioactivity and Regenerative Potential of Calcium Silicate-Based Endodontic Sealers on Human Periodontal Ligament Stem Cells: A Systematic Review of In Vitro Cellular Mechanisms

👤 Ali M Falatah
📅 2026-10-23 📖 1 min 📝 63 words

LOW VISION

Summary: Calcium silicate-based endodontic sealers are bioactive obturation materials that may support periapical regeneration, yet in vitro evidence remains fragmented and methodologically heterogeneous. This review evaluated whether these sealers show consistent cellular bioactivity in human periodontal li...

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Rehabilitation for Children With Cancer: Global Opportunities and Need for Systems-Aligned Interventions

Daniel C Moreira 2026-10-23 1 min 33 words LOW VISION

Summary: CONCLUSIONS: This survey highlights opportunities to improve access to rehabilitation and assistive products through strengthening referral pathways, workforce capacity, financing, care coordination, and monitoring. These components are interdependent and require coordinated system-level action.

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Action features dominate cortical representation during natural vision

Samuel A Nastase 2026-10-23 1 min 70 words LOW VISION

Summary: Cortical resources are allocated into systems specialized for processing ecologically relevant features of the world. These features are typically studied in isolation using controlled experimental stimuli, making it difficult to assess the relative importance of different kinds of features that ten...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

Lisa M McTeague 2026-10-20 1 min 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

Aliaa Elabd 2026-10-20 1 min 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

👤 Peng Chen 📅 2026-10-23 📖 1 min 📝 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

👤 Xiaojun Zhou 📅 2026-10-23 📖 1 min 📝 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

Fanshuo Zeng
2026-10-21 1 min 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

Xuqun You
2026-10-22 1 min 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi
📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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Multimodal Analysis of Brain Network Abnormalities in Autism Spectrum Disorder: a Graph Neural Network-based Fusion Investigation of EEG and fNIRS

Rihui Li 2026-10-23 1 min 24 words FNIRS

Summary: CONCLUSION: The proposed framework effectively captures quantitative insights ranging from specific connection-level variations to network-level differences, identifying candidate discriminative connectivity patterns within this cohort.

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Towards semantic brain-computer interfacing: The graded inventory of semantic triggers (GIST) paradigm

Ian Daly 2026-10-22 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Semantic Brain-Computer Interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual processes an...

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Preclinical models of Wilson's disease: From pathogenesis to clinical translation

Zhi-Ying Wu 2026-10-22 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Wilson's disease (WD) is an autosomal recessive disorder of copper homeostasis caused by pathogenic ATP7B variants, leading predominantly to hepatic and neurological injury. Extensive heterogeneity in genotype, age at onset, clinical presentation, and treatment response complicates both diagnosis an...

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Discussion on the closed-loop regulation strategy for acupuncture "Deqi" based on brain-computer interface and its peripheral-central mechanisms

Rong Luo 2026-10-22 1 min 72 words BRAIN COMPUTER INTERFACE

Summary: It is held that "Deqi" is the key to the therapeutic effect of acupuncture, but its evaluation has long relied on subjective feelings and lacks objective biological indicators. In the present article, we systematically review the current research status of brain-computer interface (BCI) technology b...

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RLAgent: retrieval-augmented LLM agents for code-free RNA region-ligand interaction modeling

Zhe Liu 2026-10-23 1 min 61 words BRAIN COMPUTER INTERFACE

Summary: Accurately modeling RNA-ligand interactions is essential for understanding RNA function and for designing RNA-targeted therapeutics. Existing computational approaches mainly focus on predicting either full-length RNA structures or discrete binding sites, lacking explicit modeling of RNA regions that...

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Graded multisensory neurofeedback via the iNems system for comprehensive upper limb recovery after stroke: a multiple baseline study

Kiichiro Morita 2026-10-23 1 min 37 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: This first MBD-validated demonstration shows that graded, multisensory graded neurofeedback based on ERD magnitude can improve motor, self-reported numbness, and cognitive domains after stroke. These findings support adaptive closed-loop therapy as a promising, multifaceted rehabilitatio...

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Conductive hydrogel bioelectronics for chronic neuromodulation: From biomimetic design and bioactive interfaces to integrated neural platforms

Xulin Hu 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: In biomedical engineering, hydrogel bioelectronics brings together the interface between flexible electronic devices and biological systems, opening up a new way for chronic neural regulation. Here we focus on the recent progress in this field, from material design to system integration. The design ...

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Adaptive local temporal weighting for short-window EEG-based SSVEP recognition using FB-eTRCA

Zehua Wang 2026-10-23 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: To improve short-window SSVEP recognition, we integrated adaptive temporal local weighting (ATLW) with filter-bank ensemble task-related component analysis (FB-eTRCA). ATLW estimates local discriminability from training trials, converts window scores into sample weights, and applies the training-der...

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Clinically unresponsive patients with acute brain injury activate a brain-computer interface

Jan Claassen 2026-10-23 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: Brain activation to verbal commands can be detected by EEG in clinically unresponsive patients. These patients with cognitive motor dissociation (CMD) are unable to communicate with the environment despite preserved consciousness. Advanced brain-computer interfaces (BCI) are promising to support mot...

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Decoding motor imagery EEG signals by a deep learning network integrating multiscale filtering and temporal-channel attention mechanisms

Qingguo Wei 2026-10-23 1 min 75 words BRAIN COMPUTER INTERFACE

Summary: Due to their low signal-to-noise ratio and non-stationarity, accurately classifying EEG signals is still one of the main hindrances that limit the real-world applications of a brain-computer interface (BCI). Deep learning (DL) is a potential method for addressing the problem. However, how to create ...

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Biophysically constrained learning from healthy neural priors improves motor decoding in pathological cortical networks

Jiayuan He 2026-10-23 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: High-performance brain-computer interfaces (BCIs) rely on stable neural representations, yet brain functional impairments inherently disrupt canonical encoding, creating a fundamental decoding paradox. Conventional data-driven approaches struggle to bridge this gap because of pathological data scarc...

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Saturday Daily Thread: Resource Request and Sharing! Daily Thread

/u/ AutoModerator 2026-10-24 1 min 180 words REDDIT PYTHON

Summary: <!-- SC_OFF --> <h1>Weekly Thread: Resource Request and Sharing 📚</h1> Stumbled upon a useful Python resource? Or are you looking for a guide on a specific topic? Welcome to the Resource Request and Sharing thread! <h2>How it Works:</h2> Request: Can't...

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https://www.reddit.com/r/Python/comments/1xdu1e2/saturday_daily_thread_resource_request_and/

German court calls Google aerial imagery 'spying'

👤 rdmuser 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.courthousenews.com/german-court-calls-google-aerial-imagery-spying/ Comments URL: <a href="https://news.ycombinator.com/item?id=50216142">https://news.ycombinator.com/item...

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Evolving Threat: http2shell is an RCE in node, Python, and more

👤 cowlevel 📅 2026-10-23 📖 1 min 📝 13 words HACKER NEWS

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Recording and slides now available for ERPLAB Studio webinar

Steve Luck | 2024-06-28 | 1 min | 30 words | ERP BOOT CAMP

Summary: We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

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https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

2025-12-02 | 1 min | 166 words | REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

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Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11361153

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IEEE Engineering in Medicine and Biology Society

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The Concise Language Paradigm in left and right hemisphere stroke: a longitudinal source-localized EEG study

👤 Roos, N. M., Oostenveld, R., Piastra, M. C., Tuladhar, A. M., Ruiter, M. B., Kessels, R. P. C., Westner, B. U., Piai, V.

📅 2026-10-23 📖 1 min 📝 249 words BIORXIV NEUROSCIENCE

Summary: The mechanisms underlying language recovery after stroke remain poorly understood. Most previous studies have either focused on language comprehension or production and/or relied on hemodynamic measures, with limited complementary electrophysiological evidence. Here, we combined the Concise Language...

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Prefrontal cortical morphometric covariance and cognitive task performance in healthy young adults: an exploratory analysis of Human Connectome Project data

Roman Chivers 2026-10-22 1 min 343 words FRONTIERS HUMAN NEUROSCIENCE

Summary: The prefrontal cortex (PFC) is implicated in many cognitive tasks, including working memory, reward-based decision-making, executive control, and metacognition. This exploratory analysis investigated whether individual expression of PFC cortical thickness and surface area covariance components is as...

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-10-22 1 min 231 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMotor imagery-based brain-computer interface (MI-BCI) holds great potential in neurorehabilitation and motor assistance. Data quality and motor intention decoding effectiveness are closely associated with participants' engagement. However, previous MI tasks ignored active engagement regula...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1908608

Explainable fMRI decoding reveals neurobiological signatures of emotional arousal complementary to traditional analyses

👤 Daniel Agostinho, Davide Borra, Miguel Castelo-Branco and Marco Simões

📅 2026-10-13 📖 1 min

📝 283 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Understanding how emotional arousal is represented in the brain is fundamental for elucidating affective regulation and its alterations in autism spectrum disorder. Classical neuroimaging approaches, such as the general linear model (GLM), have provided important insights into emotional p...

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https://iopscience.iop.org/article/10.1088/1741-2552/aead27

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

> **Summary:** The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ex vivo evaluation of corneal permeation and irritation potential associated with proparacaine hydrochloride misuse

👤 Yolima Baena 📅 2026-10-20 📖 1 min 📝 35 words LOW VISION

Summary: CONCLUSION: Proparacaine demonstrates dose-dependent permeation and histopathological changes, emphasizing the risks associated with misuse. The ex vivo model provides relevant and ethically aligned evidence to support ocular hazard assessment through the characterizing of histopathological changes.

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Shared variance across FrACT visual tasks: Robust positive associations but inconclusive latent structure

👤 Luca Battaglini 📅 2026-10-21 📖 1 min 📝 68 words LOW VISION

Summary: A long-standing question in vision science is whether individual differences in visual performance reflect a general visual ability or a set of largely task-specific processes. Previous studies have often reported weak associations across heterogeneous perceptual tasks, challenging the idea of a com...

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New challenges in horticultural IPM: artificial intelligence can enhance the use of biocontrol agents

👤 Andrea Lovazzano 📅 2026-10-22 📖 1 min 📝 59 words LOW VISION

Summary: Modern horticulture faces the dual imperatives of increasing global food production while mitigating the severe environmental and health impacts of conventional chemical pesticides. Biological control agents (BCAs), which utilize beneficial microorganisms, macroorganisms, semiochemical and botanical...

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Bridging the performance-efficiency gap: a resource-efficient hybrid DL-ML framework for jackfruit disease detection

👤 Hezerul Bin Abdul Karim
📅 2026-10-22 📖 1 min 📝 62 words LOW VISION

Summary: Complex environment variability and limited computational resources in agricultural settings create challenges in automated detection in tropical regions, where jackfruit diseases pose major threats to harvest productivity and food security. Conventional machine learning models often fail to general...

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Atomic Defect-Mediated Charge Trapping Enables Multimodal Plasticity in van der Waals Heterostructures for In-Sensor Neuromorphic Vision

👤 Zhiqun Lin
📅 2026-10-22 📖 1 min 📝 60 words LOW VISION

Summary: Neuromorphic hardware that unifies sensing, memory, and learning at the device level remains challenging due to limited optoelectronic comodulation and rigid architectures. Here, we report a flexible, van der Waals (vdW)-integrated MoS2/hexagonal boron nitride (h-BN)/graphene memtransistor in which ...

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Low-Dimensional Chiral Perovskites for Polarization-Sensitive Photodetection

Zhiyong Fan 2026-10-22 1 min 57 words LOW VISION

Summary: Emerging low-dimensional chiral perovskites have recently demonstrated significant application potential in circularly polarized light detection, spintronics, and quantum information technology due to their unique chiroptical activity and spin-selective properties. This review systematically summari...

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Visual context influences lateral balance while walking on winding paths

Jonathan B Dingwell 2026-10-22 1 min 28 words LOW VISION

Summary: BACKGROUND: Navigating daily environments necessitates adapting to ever-changing conditions while maintaining stability. This requires integrating sensory feedback, primarily from vision, to maintain balance and maneuverability as we walk.

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Bioactivity and Regenerative Potential of Calcium Silicate-Based Endodontic Sealers on Human Periodontal Ligament Stem Cells: A Systematic Review of In Vitro Cellular Mechanisms

👤 Ali M Falatah
📅 2026-10-23 📖 1 min 📝 63 words

LOW VISION

Summary: Calcium silicate-based endodontic sealers are bioactive obturation materials that may support periapical regeneration, yet in vitro evidence remains fragmented and methodologically heterogeneous. This review evaluated whether these sealers show consistent cellular bioactivity in human periodontal li...

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Rehabilitation for Children With Cancer: Global Opportunities and Need for Systems-Aligned Interventions

Daniel C Moreira 2026-10-23 1 min 33 words LOW VISION

Summary: CONCLUSIONS: This survey highlights opportunities to improve access to rehabilitation and assistive products through strengthening referral pathways, workforce capacity, financing, care coordination, and monitoring. These components are interdependent and require coordinated system-level action.

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Action features dominate cortical representation during natural vision

Samuel A Nastase 2026-10-23 1 min 70 words LOW VISION

Summary: Cortical resources are allocated into systems specialized for processing ecologically relevant features of the world. These features are typically studied in isolation using controlled experimental stimuli, making it difficult to assess the relative importance of different kinds of features that ten...

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Altered abdominal spatial mapping despite preserved tactile acuity in adolescents with Restrictive Eating Disorders

Martina M Mensi 2026-06-25 1 min 65 words TACTILE ACUITY

Summary: Disturbances in body perception are a consistent feature of Restrictive Eating Disorders (REDs). In adult patients diagnosed with REDs, body dissatisfaction and overestimation of body weight have been associated with altered tactile acuity and biases in proprioceptive localization. Whether similar a...

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Immersive virtual reality with synchronous neurostimulation for upper-limb recovery after stroke: a randomized feasibility trial

Stanisa Raspopovic 2026-06-26 1 min 60 words TACTILE ACUITY

Summary: Stroke affects 15 million people annually and leaves 5 million permanently disabled. In the chronic phase (>3 months after stroke), patients often experience persistent sensorimotor deficits and altered body representation, yet rehabilitation delivery remains partial and inconsistent, highlighting a...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

👤 Martin E Héroux
📅 2026-07-18 📖 1 min 📝 69 words

TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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In vitro carbapenem activity in Neisseria gonorrhoeae

Béatrice Berçot 2026-09-04 1 min 60 words BRAILLE

Summary: CONCLUSIONS: Carbapenem MICs remain low in N. gonorrhoeae isolates but are higher in those with mosaic penA alleles than in those with non-mosaic penA alleles. MICs for ertapenem are lower than those for other carbapenems, supporting its use for the treatment of infections with ceftriaxone-resistant...

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Multi-Modal Low-Power Adaptive Braille Recognition System Based on Hf(0.3)Zr(0.7)O(2) Memristor

Xiaobing Yan 2026-09-09 1 min 57 words BRAILLE

Summary: Tactile-assisted perception systems hold great promise for robotics and human-machine interaction, yet current assistive systems for visually impaired individuals suffer from redundant data transmission, complex fusion algorithms, and high-power consumption. These limitations can introduce milliseco...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

Priyanga Chandrasekaran
2026-09-22 1 min 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Comparative chloroplast genomics of Ajania sericea and sympatric Asteraceae species identifies shared candidate molecular features associated with alpine adaptation

Xi Chen 2026-10-20 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: This study showed that the chloroplast genomes of four sympatric Asteraceae species were highly conserved in overall structure, but still exhibited certain differences in gene loss, IR boundary variation, repeat sequence composition, and local hypervariable regions. These features provi...

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Brain Stimulation as a Treatment for Apathy: A Scoping Review of Effects and Safety

👤 Lisa M McTeague 📅 2026-10-20 📖 1 min 📝 46 words TDCS TACS TRNS

Summary: CONCLUSIONS: Investigations into brain stimulation as a treatment for apathy show some promise. Conclusions were limited by small sample sizes, heterogeneous study designs, and variable outcome measures. Randomized trials with common validated apathy instruments, standardized stimulation parameters,...

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Effect of Different Cortical Targets of Transcranial Magnetic Stimulation on Freezing of Gait in Parkinsonism: A Systematic Review

👤 Aliaa Elabd 📅 2026-10-20 📖 1 min 📝 68 words TDCS TACS TRNS

Summary: BACKGROUND AND PURPOSE: Freezing of gait (FOG) is a disabling and episodic gait disturbance in patients with Parkinson's, with varying responses to noninvasive brain stimulation. Transcranial magnetic stimulation (TMS) has been explored as a therapeutic option; however, the optimal cortical target f...

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Sleep Spindles and Slow Oscillations as Potential Biomarkers and Therapeutic Targets in Mild-to-Moderate Alzheimer's Disease

👤 Thien Thanh Dang-Vu 📅 2026-10-20 📖 1 min 📝 35 words TDCS TACS TRNS

Summary: PURPOSE OF REVIEW: This review critically evaluates evidence on sleep spindles (SP), slow oscillations (SOs), and their coordination in mild-to-moderate Alzheimer's disease (AD), focusing on their relationships with cognition, AD biomarkers, and their therapeutic potential.

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Affinity Engineering Modulates Soluble Antigen Degrader Activity

👤 Xin Zhou 📅 2026-10-20 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Soluble extracellular proteins drive many physiological and pathological signaling pathways, and targeted degradation offers a potential strategy for their therapeutic modulation. Here, we engineered bispecific targeting chimeras (TACs) that couple soluble proteins to constitutively internalizing ce...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Multiple Sessions of Transcranial Direct Current Stimulation Combined With Squat Training Improve Lower-Limb Maximum Strength in Healthy Adult Men

👤 Fengxue Qi
📅 2026-10-22 📖 1 min 📝 79 words

TDCS TACS TRNS

Summary: Yan, X, Zuo, Z, Nitsche, MA, Zhang, X, and Qi, F. Multiple sessions of transcranial direct current stimulation combined with squat training improve lower-limb maximum strength in healthy adult men. J Strength Cond Res XX(X): 000-000, 2026-This study aimed to investigate the effects of transcranial d...

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Time-aware frame-aligned diffusion-inspired model for late dynamic PET frame prediction from early [¹¹C] raclopride

👤 Jae Sung Lee
📅 2026-10-22 📖 1 min 📝 42 words

TDCS TACS TRNS

Summary: OBJECTIVE: To reduce the burden of dynamic PET acquisition while preserving quantitative analysis, we propose a time-aware frame-aligned diffusion-inspired residual denoising framework for predicting late dynamic PET frames from a short early acquisition window (10 min in this study) of [¹¹C] raclop...

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Effects of transcranial direct current stimulation on patient-reported outcomes in chronic low back pain: a systematic review and three-level meta-analysis

👤 Peng Chen 📅 2026-10-23 📖 1 min 📝 34 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence suggests that tDCS may reduce pain and produce a small improvement in disability in individuals with CLBP. Given the low-to-moderate certainty of the available evidence, these findings should be interpreted cautiously.

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tACS suppresses M1 polarization of microglia after spinal cord injury through the cGMP-PKG signaling pathway

👤 Xiaojun Zhou 📅 2026-10-23 📖 1 min 📝 38 words TDCS TACS TRNS

Summary: CONCLUSION: The activity of the cGMP-PKG signaling pathway is upregulated by tACS, thereby modulating microglial polarization, suppressing neuroinflammation and protecting neurons from inflammatory injury. As a non-invasive neuromodulation technique, tACS holds potential as a therapeutic approach fo...

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Neural dysfunction underlying divergent thinking and its shared mechanisms with protracted withdrawal symptoms in heroin use disorder

👤 Haijun Duan
📅 2026-10-19 📖 1 min 📝 55 words FNIRS

Summary: CONCLUSIONS: These findings reveal a sustained compensatory pathological mechanism and an imbalance in neural resource allocation during complex cognition in HUD individuals. The identified signatures of cortical activity by fNIRS offer reliable biomarkers for the rapid detection of both higher-orde...

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Functional near-infrared spectroscopy as a tool for monitoring treatment-related neurofunctional changes in posttraumatic stress disorder and trauma-related symptomatology: a narrative review

👤 Maria Cotelli
📅 2026-10-20 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Current evidence suggests that fNIRS represents a promising and clinically feasible approach for monitoring treatment-related neurofunctional changes in PTSD and trauma-related symptomatology, particularly during non-pharmacological interventions targeting prefrontal regulatory processe...

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Transcutaneous vagus nerve stimulation during semantic verbal fluency tasks: a high-density diffuse optical tomography study

👤 Li Su 📅 2026-10-20 📖 1 min 📝 55 words FNIRS

Summary: INTRODUCTION: Transcutaneous auricular vagus nerve stimulation (taVNS) is a promising adjunct for post-stroke language recovery. There are no established biomarkers for the neuromodulatory effects of taVNS. High-density diffuse optical tomography (HD-DOT) offers a non-invasive marker of task-related...

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EEG-defined differences in functional networks following beta-band transcranial alternating current stimulation in patients with disorders of consciousness: a pilot study

👤 Fanshuo Zeng
📅 2026-10-21 📖 1 min 📝 53 words FNIRS

Summary: CONCLUSIONS: Frontal beta-tACS was associated with state-dependent changes in functional network organization in DOC, with dynamic fNIRS measures that may be useful for tracking these changes. These findings provide preliminary descriptive evidence of short-term network changes associated with beta-...

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Resting-State and Task-State Dual-Paradigm fNIRS Study Reveals Dynamic Alterations in Brain Network Activity in Alzheimer's Disease With Behavioural and Psychological Symptoms

👤 Hua Hu
📅 2026-10-21 📖 1 min 📝 47 words FNIRS

Summary: CONCLUSIONS: AD-BPSD was associated with altered brain functional patterns across resting and task states, involving sensorimotor and executive networks as well as altered interhemispheric coordination. These fNIRS-derived alterations provide exploratory evidence of altered brain functional patterns...

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Task load and phase relevance modulate primary flight display change blindness: evidence from flight simulator, eye-tracking, and fNIRS

👤 Xuqun You
📅 2026-10-22 📖 1 min 📝 70 words FNIRS

Summary: Change blindness (CB) during Primary Flight Display (PFD) monitoring threatens aviation safety, yet the roles of flight phase and parameter relevance remain unclear. This study examined cognitive-neural correlates of PFD change blindness in experienced commercial pilots. Experiment 1 (N = 14) found ...

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Spatiotemporal feature-based investigation of major depressive disorder and emotion recognition using functional near infrared spectroscopy

👤 Qiuxia Shi 📅 2026-10-22 📖 1 min 📝 63 words FNIRS

Summary: Major Depressive Disorder (MDD) remains one of the most prevalent psychiatric conditions worldwide, with a marked increase in incidence and severity following the COVID-19 pandemic. Despite its rising burden, clinical diagnosis continues to rely on subjective assessments, lacking objective and repro...

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A Sensitive BLE-Enabled Wearable fNIRS Biosensor for Real-Time Monitoring and Mapping of Task-Evoked Oxygen Influxes During On-Screen Cognitive Tasks

Balaji Jayaprakash 2026-10-22 1 min 34 words FNIRS

Summary: OBJECTIVE: To develop and validate a compact Bluetooth Low Energy (BLE)-enabled wearable functional near-infrared spectroscopy (fNIRS) device for real-time hemodynamic monitoring and proposes a task-evoked metric for identifying transient oxygenation events during cognitive engagement.

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FroHyNet: frozen temporal priors and trainable hypergraph fusion for brain time-series classification

Jing Zhang 2026-10-23 1 min 51 words FNIRS

Summary: BACKGROUND: Functional brain time-series classification is constrained by small clinical cohorts. Handcrafted features are sample-efficient but discard temporal detail; end-to-end networks must estimate many temporal parameters from limited labels; and frozen time-series foundation models reduce tha...

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